

A computer-assisted proof of the bound $\Lambda \leq 0.1787854$ for the de Bruijn–Newman constant

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Abstract

We prove that the de Bruijn–Newman constant satisfies

$$\Lambda \leq \frac{893927}{5000000} = 0.1787854$$

by applying the Polymath 15 barrier criterion at the exact parameters $X = 6000000185827$, $t_0 = 129/800$, and $y_0^2 = 87677/2500000$. The verified-height hypothesis follows from the theorem of Platt and Trudgian. We establish the required nonvanishing of H_{t_0} by a finite interval computation followed by analytic estimates for the remaining range. For the barrier, a further interval computation follows the boundary of a rectangle through 883 time intervals and proves that its winding number remains zero. The programs used for the finite calculations are linked where they enter the proof and described in Appendix A. Since the resulting upper bound is positive, it does not imply the Riemann hypothesis.

Human–AI collaboration disclosure. This paper was developed through an iterative collaboration between Jude Gomila and generative AI systems, with repeated cycles of mathematical exploration, program development, testing, criticism, and revision. Jude Gomila directed the work and accepts responsibility for the arguments, computations, programs, and final text. Further details appear at the end of the Introduction.

1 Introduction

1.1 The de Bruijn–Newman constant

Let Φ denote the super-exponentially decaying function

$$\Phi(u) = \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) \exp(-\pi n^2 e^{4u}),$$

and for $t \in \mathbb{R}$ define the entire functions

$$H_t(z) = \int_0^{\infty} e^{tu^2} \Phi(u) \cos(zu) du. \tag{1}$$

The function H_0 is a normalization of the Riemann ξ -function. With

$$\xi(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

analytic continuation of the Fourier identity gives, for every $z \in \mathbb{C}$, $H_0(z) = \frac{1}{8} \xi(\frac{1}{2} + \frac{iz}{2})$, so that the Riemann hypothesis (RH) is the statement that all zeros of H_0 are real. De Bruijn [1] proved that H_t has only real zeros for $t \geq \frac{1}{2}$. Newman [6] then proved that there is a finite constant $\Lambda \leq \frac{1}{2}$, the *de Bruijn–Newman constant*, such that H_t has only real zeros if and only if $t \geq \Lambda$. RH is equivalent to $\Lambda \leq 0$.

Newman conjectured $\Lambda \geq 0$, which was proved by Rodgers and Tao [9]. On the other side, de Bruijn’s classical upper bound $\Lambda \leq 1/2$, its strict improvement by Ki, Kim and Lee [4], and subsequent numerical work led to the bound

$$\Lambda \leq 0.22 \tag{2}$$

of the Polymath 15 project [8]. Subsequently Platt and Trudgian observed [7, Corollary 2] that their verification of RH to height $3 \cdot 10^{12}$ (Section 3 below), combined with the second row of the parameter table in [8, Table 1, p. 65] (which requires verified height $H > 2.51 \cdot 10^{12}$), yields the bound

$$\Lambda \leq 0.2, \tag{3}$$

which is the relevant peer-reviewed benchmark for the present argument. Platt and Trudgian further note that the next entry of the same table would give $\Lambda < 0.19$, but only conditionally on a verified height slightly above 10^{13} , beyond their theorem. The value (5) lies below both 0.2 and 0.19 while consuming only the established $3 \cdot 10^{12}$ verification height. A recent Zenodo preprint of Mosaic Intelligence [5] claims the intermediate value $\Lambda \leq 0.1875$; it has not been peer reviewed. The value examined here is numerically smaller than both 0.2 and 0.1875.

1.2 Main result

Theorem 1.1. *The de Bruijn–Newman constant satisfies*

$$\Lambda \leq \frac{893927}{5000000} = 0.1787854.$$

The proof instantiates the barrier criterion of [8, Theorem 1.2] at the exact parameters

$$X = 6000000185827, \quad t_0 = \frac{129}{800} = 0.16125, \quad y_0^2 = \frac{87677}{2500000} = 0.0350708, \tag{4}$$

with $y_0 > 0$, so that

$$t_0 + \frac{y_0^2}{2} = \frac{893927}{5000000} = 0.1787854. \tag{5}$$

One checks the elementary inequalities $0 < t_0 < \frac{1}{2}$, $0 < y_0^2 < 1 - 2t_0 = \frac{271}{400}$, and $y_0^2 + 2t_0 = \frac{893927}{2500000} < 1$, which place the parameters in the domain of the criterion.

1.3 Outline of the proof

The argument has four parts.

1. Section 2 states the Polymath 15 criterion and the effective approximation to which we apply it.
2. Section 3 obtains the required verified height from the theorem of Platt and Trudgian.

3. Sections 4 and 5 prove the required nonvanishing of H_{t_0} , first on a finite range and then on the infinite tail.
4. Section 6 proves the boundary nonvanishing needed for the barrier argument, and Section 7 completes the proof.

The finite parts of the proof use interval arithmetic. At each such point we give the exact domain, the mathematical quantity evaluated, the strict inequality returned by the program, and a link to the source. The programs use exact integer or rational inputs and outward-rounded FLINT/Arb balls [3]; Appendix A collects the links and explains how the computations cover the required domains. The source code is available in the [project repository](#).

This preprint has not undergone independent peer review.

Use of generative AI. Generative AI systems, including Anthropic Claude and OpenAI ChatGPT/Codex, were used in repeated, human-directed cycles of mathematical exploration, program development, testing, criticism, and manuscript revision. Jude Gomila selected the arguments and computations included here and accepts responsibility for the mathematical claims, the programs, and the final text.

2 The Polymath criterion and effective approximation

2.1 The barrier criterion

The logical skeleton of the proof is the following criterion.

Theorem 2.1 (Polymath criterion, specialized from [8, Theorem 1.2]). *Let $X > 0$, $0 < t_0 < \frac{1}{2}$ and $0 < y_0 < \sqrt{1 - 2t_0}$. Suppose that:*

- (i) (**Verified height.**) *There are no zeros $\zeta(\sigma + iT) = 0$ with*

$$\frac{1 + y_0}{2} \leq \sigma \leq 1, \quad 0 \leq T \leq \frac{X}{2}.$$

- (ii) (**Final-time right region.**) *There are no zeros $H_{t_0}(x + iy) = 0$ with*

$$x \geq X + \sqrt{1 - y_0^2}, \quad y_0 \leq y \leq \sqrt{1 - 2t_0}.$$

- (iii) (**Intermediate-time barrier.**) *There are no zeros $H_t(x + iy) = 0$ with*

$$X \leq x \leq X + \sqrt{1 - y_0^2}, \quad \sqrt{y_0^2 + 2(t_0 - t)} \leq y \leq \sqrt{1 - 2t}, \quad 0 \leq t \leq t_0.$$

Then

$$\Lambda \leq t_0 + \frac{y_0^2}{2}.$$

Sections 3, 4–5, and 6 establish hypotheses (i), (ii), and (iii), respectively, at the parameters (4).

For hypothesis (i) one uses the normalization connecting H_0 to ζ : a zero $H_0(x + iy) = 0$ gives $\xi(\frac{1-y+ix}{2}) = 0$; the functional equation $\xi(s) = \xi(1-s)$ followed by complex conjugation produces the representative zero $\xi(\frac{1+y+ix}{2}) = 0$ with $\sigma = \frac{1+y}{2} \geq \frac{1+y_0}{2}$ and ordinate $T = \frac{x}{2}$, which is the region excluded in (i). (The functional equation and the reality symmetry $\overline{\xi(\bar{s})} = \xi(s)$ are the classical inputs; the affine algebra of this “sign map” is elementary.)

2.2 The effective approximation

For $t > 0$ write, following [8],

$$g_t(z) = \frac{H_t(z)}{B_t(z)}, \quad (6)$$

where the normalizing factor is as follows. Let Log be the standard branch of the logarithm, with branch cut $(-\infty, 0]$ and imaginary part in $(-\pi, \pi]$, and on $\mathbb{C} \setminus (-\infty, 1]$ define

$$M_0(s) = \frac{\sqrt{2\pi}}{16} s(s-1) \pi^{-s/2} \exp\left(\left(\frac{s}{2} - \frac{1}{2}\right) \text{Log} \frac{s}{2} - \frac{s}{2}\right), \quad (7)$$

$$\alpha(s) = (\log M_0)'(s) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2} \text{Log} \frac{s}{2\pi}, \quad (8)$$

$$M_t(s) = \exp\left(\frac{t}{4} \alpha(s)^2\right) M_0(s), \quad B_t(x+iy) = M_t\left(\frac{1+y-ix}{2}\right). \quad (9)$$

These are equations (6), (8)–(11) of [8]. In particular M_0 , and hence M_t , takes values in $\mathbb{C} \setminus \{0\}$ on its domain; since $(1+y-ix)/2$ is non-real whenever $x \neq 0$, it avoids the branch cut. Therefore $B_t(x+iy) \neq 0$ for all $t \geq 0$, $x > 0$ and $y > 0$, as noted explicitly in [8, §1, after (11)]. This nonvanishing is used below both in the reduction of zero-freeness of H_t to lower bounds on $|f_t|$ and in the argument-principle step of the barrier argument (Section 6). Theorem 1.3 of [8] provides, in the region

$$x \geq 200, \quad 0 \leq y \leq 1, \quad 0 < t \leq \frac{1}{2}, \quad (10)$$

an approximation

$$g_t(x+iy) = f_t(x+iy) + O_{\leq}(e_A + e_B + e_{C,0}), \quad (11)$$

where $O_{\leq}(E)$ denotes a quantity of absolute value at most E , the main term is the finite sum (equation (14) of [8])

$$f_t(x+iy) = \sum_{n=1}^N b_t(n) n^{-s_*} + \gamma \sum_{n=1}^N n^y b_t(n) n^{-\overline{s_*} - \kappa}, \quad b_t(n) = \exp\left(\frac{t}{4} \log^2 n\right), \quad (12)$$

(we write $b_t(n)$ for the quantity denoted b_n^t in [8]; all other symbols in (12) agree with the notation there, including the placement of κ alongside the conjugated exponent $\overline{s_*}$ in the second sum) with:

- the window index $N = \lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \rfloor$;
- the exact quantities (equations (16)–(18) of [8])

$$\gamma = \frac{M_t\left(\frac{1-y+ix}{2}\right)}{M_t\left(\frac{1+y-ix}{2}\right)}, \quad (13)$$

$$s_* = \frac{1+y-ix}{2} + \frac{t}{2} \alpha\left(\frac{1+y-ix}{2}\right), \quad (14)$$

$$\kappa = \frac{t}{2} \left[\alpha\left(\frac{1-y+ix}{2}\right) - \alpha\left(\frac{1+y+ix}{2}\right) \right], \quad (15)$$

which satisfy, by equations (20)–(22) of [8], the bounds

$$|\gamma| \leq e^{0.02y} \left(\frac{x}{4\pi} \right)^{-y/2}, \quad (16)$$

$$\operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t}{4} \log \frac{x}{4\pi} - \frac{t}{2x^2} \left(1 - 3y + \frac{4y(1+y)}{x^2} \right)_+, \quad (17)$$

$$|\kappa| \leq \frac{ty}{2(x-6)}; \quad (18)$$

- and error terms $e_A, e_B, e_{C,0}$ with explicit majorants (equations (71)–(72) and (23)–(24) of [8]).

Since $B_t \neq 0$, nonvanishing of H_t on a region follows from the strict inequality $|f_t| > e_A + e_B + e_{C,0}$ there; this implication is itself stated in [8] as Corollary 1.4 (“Criterion for non-vanishing”).

2.3 The conservative error constant 10.50

Our proof deliberately does *not* use the constant 10.44 displayed in equation (24) of [8] for the majorization of $e_{C,0}$. Instead it derives a slightly weaker bound directly from Proposition 6.6(vi) of [8], which contains the expression

$$\exp \left(\frac{3 \left| \log \frac{x}{4\pi} + \frac{i\pi}{2} \right| + 3.58}{x - 8.52} \right) \left(1 + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{6.92}{x - 12} \right).$$

On the theorem domain $x \geq 200$ one has $x - 8.52 > x - 12 > 0$; applying $1 + u \leq e^u$ to the second factor and replacing the denominator $x - 8.52$ by the smaller denominator $x - 12$ yields the *conservative corollary*

$$e_{C,0} \leq \left(\frac{x}{4\pi} \right)^{-(1+y)/4} \exp \left(-\frac{t}{16} \log^2 \frac{x}{4\pi} + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{3 \left| \log \frac{x}{4\pi} + \frac{i\pi}{2} \right| + 10.50}{x - 12} \right), \quad (19)$$

with $3.58 + 6.92 = \frac{21}{2} = 10.50$ exactly. Every numerical error computation below (finite range, tail, and barrier) uses (19), and uses the denominator $x - 6.66$, which appears in equation (23) and in Proposition 6.6(iv),(v) of [8], for $e_A + e_B$.

Remark 2.2. Equation (24) of [8] displays the bound (19) with the constant 10.44 in place of 10.50. The sentence following Proposition 6.6 in [8] states that (24) is obtained from part (vi) of the proposition by “the inequality $1 + u \leq \exp(u)$ ” followed by the bound $\frac{1}{x-8.52} \leq \frac{1}{x-12}$; carrying out this recipe yields the constant $3.58 + 6.92 = 10.50$, and we are not aware of an argument in [8] producing the displayed 10.44. Accordingly, our proof does not rely on (24): all error computations use only (19), which follows rigorously from Proposition 6.6(vi) by the two elementary steps just described, uniformly on the domain (10). (The weaker constant costs a factor $e^{0.06/(x-12)}$, which is utterly negligible at $x \approx 6 \times 10^{12}$.)

3 Hypothesis (i): the verified height

Platt and Trudgian [7, Theorem 1] proved that every nontrivial zero of ζ with ordinate in $(0, T_{\text{PT}}]$,

$$T_{\text{PT}} = 3000175332800,$$

lies on the critical line; their computation accounts for the lowest 12 363 153 437 138 nontrivial zeros. Counting sign changes of the completed zeta function on the critical line and using Turing's method to confirm completeness yields the off-line exclusion form used here. Hypothesis (i) of Theorem 2.1 requires the absence of zeros with $\frac{1+y_0}{2} \leq \sigma \leq 1$ and $0 \leq T \leq X/2$. Since

$$\frac{X}{2} = 3000000092913.5, \quad T_{\text{PT}} - \frac{X}{2} = \frac{350479773}{2} > 0,$$

the verified region covers all ordinates $0 < T \leq X/2$ with margin $\approx 1.75 \times 10^8$. On the segment $T = 0$, $\frac{1}{2} < \sigma < 1$, one argues directly: the Dirichlet eta function satisfies $\eta(\sigma) = \sum_{n \geq 1} (-1)^{n-1} n^{-\sigma} > 0$ for $0 < \sigma < 1$ (group consecutive terms), while $\eta(\sigma) = (1 - 2^{1-\sigma})\zeta(\sigma)$ with $1 - 2^{1-\sigma} < 0$; hence $\zeta(\sigma) < 0 \neq 0$ there. At $\sigma = 1$, ζ has a pole. This supplies hypothesis (i).

We use the published Platt–Trudgian theorem as an input.

4 Hypothesis (ii), finite range: the window scan

Hypothesis (ii) concerns nonvanishing of H_{t_0} on

$$x \geq x_* := X + \sqrt{1 - y_0^2}, \quad y_0 \leq y \leq \sqrt{1 - 2t_0}. \quad (20)$$

The proof splits this region at the abscissa corresponding to the window index $N_* = 3840000$: the finite range $690988 \leq N \leq 3840000$ is handled in this section by 3,149,013 individual interval computations, and the infinite range $N \geq 3840000$ in Section 5 by a single analytic contraction argument. The two ranges overlap on a full window, eliminating endpoint risk.

4.1 Windows and the freeze lemma

At $t = t_0$ define

$$x_N = 4\pi \left(N^2 - \frac{t_0}{16} \right), \quad W_N = [x_N, x_{N+1}),$$

so that $x \in W_N$ iff the window index of x equals N . Exact rational Machin-type bounds for π and exact integer square-root brackets give

$$x_{690988} < x_* < x_{690989}, \quad (21)$$

with margins exceeding 5.37×10^6 and 1.19×10^7 respectively; thus x_* lies strictly inside W_{690988} , and the half-open windows tile $[x_*, x_{3840001})$ with no gaps or overlaps.

The computations of Section 4.3 evaluate, for each N , the majorants of (16)–(18) *frozen at the left edge* $x = x_N$ of the window. The following elementary lemma justifies this.

Lemma 4.1 (Window freeze). *For $y_0 \leq y \leq \sqrt{1 - 2t_0}$ define*

$$\begin{aligned} G(x, y) &= e^{y/50} \left(\frac{x}{4\pi} \right)^{-y/2}, \\ K(x, y) &= \frac{t_0 y}{2(x - 6)}, \\ \Sigma(x, y) &= \frac{1 + y}{2} + \frac{t_0}{4} \log \frac{x}{4\pi} - \frac{t_0}{2x^2} \left(1 - 3y + \frac{4y(1 + y)}{x^2} \right)_+. \end{aligned}$$

Then for every N in the finite range, every $x \in W_N$ and every y in the stated range,

$$G(x, y) \leq G(x_N, y), \quad K(x, y) \leq K(x_N, y), \quad \Sigma(x, y) \geq \Sigma(x_N, y).$$

Proof. $\partial_x G = -\frac{y}{2x} G < 0$ and $\partial_x K = -\frac{t_0 y}{2(x-6)^2} < 0$ give the first two inequalities. For the third, write $h(x, y) = 1 - 3y + A(y)/x^2$ with $A(y) = 4y(1+y) \geq 0$; then $h_x = -2A/x^3 \leq 0$, so h_+ is nonincreasing in x , and since x^{-2} is positive and decreasing, the product $x^{-2}h_+$ is nonincreasing. The logarithmic term is strictly increasing, so Σ is strictly increasing in x , including across the kink $h = 0$ (both one-sided derivatives there are positive). $\square$

Note that $e^{y/50} = e^{0.02y}$, matching (16); G, K, Σ frozen at x_N are precisely the quantities g_N (bound for $|\gamma|$), k_N (bound for $|\kappa|$) and σ_N (lower bound for $\operatorname{Re} s_*$) used in the definitions of Section 4.3.

4.2 The mollified lower bound for $|f_{t_0}|$

Fix a window N , a height $y > 0$, and a finite set of auxiliary primes $\mathcal{P}$ with $D = \prod_{p \in \mathcal{P}} p$. Let g_N, k_N, σ_N be as above, and introduce the real *Euler mollifier*

$$E(s_*) = \prod_{p \in \mathcal{P}} (1 - b_{t_0}(p) p^{-s_*}) = \sum_{d|D} \lambda_d d^{-s_*}, \quad \lambda_d = (-1)^{\omega(d)} \prod_{p|d} b_{t_0}(p) \in \mathbb{R}.$$

Define the convolved coefficients (for $1 \leq n \leq DN$)

$$B_{N,n} = \sum_{\substack{d|D, d|n \\ n/d \leq N}} \lambda_d b_{t_0}(n/d), \quad A_{N,n} = \sum_{\substack{d|D, d|n \\ n/d \leq N}} \lambda_d b_{t_0}(n/d) (n/d)^y,$$

so $B_{N,1} = A_{N,1} = 1$, and set

$$\begin{aligned} M_N &= \sum_{d|D} |\lambda_d| d^{-\sigma_N}, & C_N &= \sum_{m=2}^N b_{t_0}(m) (m^{k_N} - 1) m^{y-\sigma_N}, \\ L_N &= \frac{1 - g_N - \sum_{n=2}^{DN} (|B_{N,n}| + g_N |A_{N,n}|) n^{-\sigma_N}}{M_N} - g_N C_N. \end{aligned} \quad (22)$$

Lemma 4.2 (Mollified lower bound). *If $L_N > 0$, then $|f_{t_0}(x + iy)| \geq L_N$ for all $x \in W_N$.*

Proof. Write (12) as $f_t = A + \gamma C_\kappa$, where

$$A = \sum_{m \leq N} b_t(m) m^{-s_*}, \quad C_\kappa = \sum_{m \leq N} m^y b_t(m) m^{-\overline{s_*} - \kappa}.$$

Exact Dirichlet convolution gives the two identities

$$E(s_*) A = \sum_{n=1}^{DN} B_{N,n} n^{-s_*}, \quad \overline{E(s_*)} C_0 = \sum_{n=1}^{DN} A_{N,n} n^{-\overline{s_*}}, \quad (23)$$

where the second identity uses that the λ_d are *real* (so that $\overline{E(s_*)} = \sum_d \lambda_d d^{-\overline{s_*}}$). By the triangle inequality and $\operatorname{Re} s_* \geq \sigma_N$,

$$|EA| \geq 1 - \sum_{n=2}^{DN} |B_{N,n}| n^{-\sigma_N}, \quad |\overline{E} C_0| \leq 1 + \sum_{n=2}^{DN} |A_{N,n}| n^{-\sigma_N}.$$

For complex κ with $|\kappa| \leq k_N$,

$$|m^{-\kappa} - 1| = |e^{-\kappa \log m} - 1| \leq e^{|\kappa| \log m} - 1 \leq m^{k_N} - 1, \quad (24)$$

whence $|\overline{E}(C_\kappa - C_0)| \leq |E|C_N$. Combining, and using $|\gamma| \leq g_N$,

$$|E||f_t| \geq Q_N - g_N|E|C_N, \quad Q_N := 1 - g_N - \sum_{n=2}^{DN} (|B_{N,n}| + g_N|A_{N,n}|)n^{-\sigma_N}.$$

If $L_N > 0$ then $Q_N = M_N(L_N + g_N C_N) > 0$; the displayed inequality then forces $E \neq 0$ (otherwise $0 \geq Q_N > 0$). Dividing by $|E|$ and using $|E| \leq M_N$ together with $Q_N > 0$,

$$|f_t| \geq \frac{Q_N}{|E|} - g_N C_N \geq \frac{Q_N}{M_N} - g_N C_N = L_N. \quad \square$$

The point of Lemma 4.2 is that any per-window lower bound for L_N uses the same normalization as the additive error in (11); no further Euler-factor conversion intervenes.

4.3 The lower bounds for L_N

For each integer N with $690988 \leq N \leq 3840000$, define a set of auxiliary primes

$$\mathcal{P}(N) = \begin{cases} \{2, 3, 5, 7, 11\}, & 690988 \leq N \leq 728999, \\ \{2, 3, 5, 7\}, & 729000 \leq N \leq 818999, \\ \{2, 3, 5\}, & 819000 \leq N \leq 1027999, \\ \{2, 3\}, & 1028000 \leq N \leq 3840000, \end{cases} \quad (25)$$

and let L_N denote the quantity (22) formed with the prime set $\mathcal{P}(N)$, at $t = t_0$ and $y = y_0$, with the constants $g_N = G(x_N, y_0)$, $k_N = K(x_N, y_0)$, $\sigma_N = \Sigma(x_N, y_0)$ frozen at the window's left edge as in Lemma 4.1. (The number of terms in L_N grows with $|\mathcal{P}(N)|$; the choice (25) uses the larger, more effective mollifiers only on the ranges of N where they are needed.)

Proposition 4.3 (Lower bounds for L_N). *For every integer $690988 \leq N \leq 3840000$,*

$$L_N \geq T_{\min} = 791366 \times 10^{-12} > 0.$$

More precisely, the minimum of L_N over each of the four ranges in (25) is bounded below as listed in Table 1; the smallest certified lower bound is $T_{\min}$ and is obtained at $N = 690988$.

$\mathcal{P}(N)$	range of N	lower bound for $\min L_N$
$\{2, 3, 5, 7, 11\}$	$690988 \leq N \leq 728999$	0.000000791366
$\{2, 3, 5, 7\}$	$729000 \leq N \leq 818999$	0.000315112459
$\{2, 3, 5\}$	$819000 \leq N \leq 1027999$	0.000305788807
$\{2, 3\}$	$1028000 \leq N \leq 3840000$	0.000309285478

Table 1: The auxiliary prime sets (25) and, for each of the four ranges of N , the lower bound for $\min_N L_N$ established by the computation described in the proof of Proposition 4.3.

Proof. The proof is computer-assisted. The [finite-window program](#) computes, for every required integer N , an outward-rounded interval whose lower endpoint is a lower bound for the quantity L_N in (22). It has two evaluation modes. The first evaluates every term of (22) directly. The second groups consecutive values of N , freezes each weight at the endpoint that gives the larger majorant, expands the remaining smooth dependence in Taylor series, and adds an explicit Lagrange remainder. The monotonicity of g_N established in Lemma 4.1 makes the frozen weights conservative. Each computed lower endpoint is therefore rigorously no greater than the exact value of L_N .

The calculation uses exact rational inputs and outward-rounded FLINT/Arb arithmetic at 256 bits. The four runs use the prime sets and integer ranges in (25) and together cover every one of the 3,149,013 integers $690988 \leq N \leq 3840000$. Every computed lower endpoint is positive; the smallest certified lower bound in each range is displayed in Table 1; across all four ranges the smallest is 791366×10^{-12} , obtained for $N = 690988$. This proves the proposition. $\square$

4.4 Transfer to the full height range

Proposition 4.3 concerns the single height $y = y_0$, while hypothesis (ii) requires nonvanishing for all $y \in [y_0, \sqrt{1-2t_0}]$. For fixed N write $L_N(y)$ for the quantity (22) with its explicit y -dependence restored (the constants g_N, k_N, σ_N frozen in x as before, but evaluated at height y ; likewise the coefficients $A_{N,n}$ and the sums M_N, C_N), so that $L_N = L_N(y_0)$. Throughout this subsection $t = t_0$ and N are fixed, and we abbreviate

$$q_N = N^2 - \frac{t_0}{16}, \quad g = \frac{1}{50} - \frac{1}{2} \log q_N (< 0), \quad \rho_N = \frac{t_0}{2(x_N - 6)},$$

so that $g_N(y) = e^{yg}$ exactly and $k_N(y) = \rho_N y$. The bridge between Proposition 4.3 and the full height range is the following statement.

Proposition 4.4 (Height transfer). *For every integer $690988 \leq N \leq 3840000$ and every $y \in [y_0, \sqrt{1-2t_0}]$,*

$$L_N(y) \geq L_N(y_0).$$

The proof establishes three monotonicity statements: as y increases over $[y_0, \sqrt{1-2t_0}]$, (a) the majorant sum

$$\mathcal{F}_{N,\mathcal{P}}(y) = g_N(y) + \sum_{n=2}^{DN} (|B_{N,n}| + g_N(y)|A_{N,n}(y)|) n^{-\sigma_N(y)}$$

subtracted in (22) is nonincreasing; (b) the mollifier bound $M_N(y)$ is positive and nonincreasing; and (c) the correction term $g_N(y)C_N(y) \geq 0$ is nonincreasing. Statement (a) is the delicate one: the crude majorant obtained by discarding the negative term $g_N g|A|$ below is not monotone, and the proof must retain the cancellation between the A -side terms. We build it up in three lemmas and one computation.

Lemma 4.5 (Slope of the frozen exponent). *For $y_0 \leq y_1 \leq y_2 \leq \sqrt{1-2t_0}$,*

$$\sigma_N(y_2) - \sigma_N(y_1) \geq \frac{1}{2}(y_2 - y_1).$$

Proof. $\sigma_N(y) = \frac{1+y}{2} + \frac{t_0}{4} \log q_N - \frac{t_0}{2x_N^2} (1 - 3y + \frac{4y(1+y)}{x_N^2})_+$. The inner expression has y -derivative $-3 + \frac{4(1+2y)}{x_N^2} < 0$ (as $4(1+2y) \leq 12 < 3x_N^2$), so its positive part is nonincreasing in y and the subtracted term is nondecreasing; the explicit term $\frac{1+y}{2}$ contributes slope $\frac{1}{2}$. $\square$

Lemma 4.6 (Monotone normalizer and correction). *On $[y_0, \sqrt{1-2t_0}]$: (b) $M_N(y)$ is positive and nonincreasing; (c) $g_N(y)C_N(y)$ is nonnegative and nonincreasing.*

Proof. (b) Each term $|\lambda_d|d^{-\sigma_N(y)}$ is positive; for $d \geq 2$ it is nonincreasing by Lemma 4.5, and the $d = 1$ term is constant.

(c) Nonnegativity is clear ($m^{k_N(y)} \geq 1$). Fix $y_0 \leq y_1 < y_2 \leq \sqrt{1-2t_0}$ and put $\Delta = y_2 - y_1$. First, $g_N(y_2) = g_N(y_1)e^{g\Delta}$ exactly. Next, for each $2 \leq m \leq N$, the term $b_t(m)(m^{\rho_N y} - 1)m^{y-\sigma_N(y)}$ of C_N grows by at most the factor $e^{(\frac{1}{2} \log m + \rho_N \log m + \frac{1}{y_0})\Delta}$: the exponent $y - \sigma_N(y)$ grows by at most $\frac{1}{2}\Delta$ (Lemma 4.5); and by the mean value theorem applied to $y \mapsto \log(m^{\rho_N y} - 1)$, whose derivative is $\rho_N \log m \frac{e^z}{e^z - 1}$ with $z = \rho_N y \log m > 0$, the bound $\frac{e^z}{e^z - 1} \leq 1 + \frac{1}{z}$ gives a growth rate at most $\rho_N \log m + \frac{1}{y} \leq \rho_N \log m + \frac{1}{y_0}$. Since a sum of positive terms grows no faster than its fastest-growing term, and $m \leq N$,

$$g_N(y_2)C_N(y_2) \leq g_N(y_1)C_N(y_1) \exp\left[\left(g + \frac{1}{2} \log N + \rho_N \log N + \frac{1}{y_0}\right)\Delta\right].$$

It remains to check that the rate is negative for every N in the range. First, $g + \frac{1}{2} \log N = \frac{1}{50} - \frac{1}{2} \log \frac{q_N}{N}$ and $\frac{q_N}{N} = N - \frac{t_0}{16N} > N - 1 \geq 690987 > e^{13.44}$ (as $e^{13.44} < 442414 \times 1.5528 < 687000$), so $g + \frac{1}{2} \log N < 0.02 - 6.72$. Second, $\frac{1}{y_0} < \frac{10^7}{1872719} < 5.3399$, since $y_0^2 = \frac{350708}{10^7} > (\frac{1872719}{10^7})^2 = 0.03507076 \dots$ gives $y_0 > \frac{1872719}{10^7}$. Third, uniformly over the range, $\rho_N \log N \leq \frac{t_0 \log(3.84 \times 10^6)}{2(x_{690988} - 6)} < \frac{0.17 \times 16}{10^{13}} < 10^{-12}$. Hence the rate is at most $0.02 + 5.3399 + 10^{-12} - 6.72 < -1.36 < 0$. $\square$

Lemma 4.7 (Dini bound in divisor-cell form). *Let $2 \leq n \leq DN$, $L = \log n$, and let $\mathcal{D}_{N,n} = \{d \mid D : d \mid n, n \leq dN\}$ be the active divisor set (which does not depend on y), assumed nonempty. With $\ell_d = \log d$ and*

$$r_d(L) = \exp\left\{\frac{t_0}{4} \left(\sum_{p|d} \log^2 p + \ell_d^2 - 2L\ell_d\right)\right\},$$

$$C_0 = \sum_{d \in \mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L),$$

$$C(y) = \sum_{d \in \mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L) d^{-y},$$

$$C'(y) = - \sum_{d \in \mathcal{D}_{N,n}} (-1)^{\omega(d)} r_d(L) d^{-y} \ell_d,$$

one has $B_{N,n} = b_t(n) C_0$, $A_{N,n}(y) = b_t(n) e^{yL} C(y)$ and $\frac{d}{dy} A_{N,n}(y) = b_t(n) e^{yL} (LC(y) + C'(y))$, and the upper right Dini derivative of the corresponding summand of $\mathcal{F}_{N,\mathcal{P}}$ satisfies

$$D_y^+ \left[(|B_{N,n}| + g_N(y) |A_{N,n}(y)|) n^{-\sigma_N(y)} \right] \leq b_t(n) n^{-\sigma_N(y)} \left(G h - \frac{L}{2} |C_0| \right),$$

where $G = G(y) = e^{y(g+L)}$ and $h = h(y) = |LC(y) + C'(y)| + (g - \frac{L}{2}) |C(y)|$.

Proof. The identities follow from $\lambda_d b_t(n/d) = (-1)^{\omega(d)} b_t(n) r_d(L)$ — indeed $\log b_t(n/d) = \frac{t_0}{4} (L - \ell_d)^2 = \log b_t(n) + \frac{t_0}{4} (\ell_d^2 - 2L\ell_d)$ and $\lambda_d = (-1)^{\omega(d)} \exp(\frac{t_0}{4} \sum_{p|d} \log^2 p)$ — together with $(n/d)^y = e^{yL} d^{-y}$; note also $g_N(y) e^{yL} = e^{y(g+L)} = G$. Since $A = A_{N,n}(y)$ is real-analytic in y ,

$$D_y^+ \{g_N |A|\} = \begin{cases} g_N(g|A| + \operatorname{sgn}(A)A'), & A \neq 0, \\ g_N |A'|, & A = 0, \end{cases} \leq g_N(g|A| + |A'|)$$

in both cases. Both factors of the summand are nonnegative and locally Lipschitz, $|B|$ is constant in y , and by Lemma 4.5, $D_y^+ n^{-\sigma_N(y)} \leq -\frac{L}{2} n^{-\sigma_N(y)}$; the product rule for upper Dini derivatives of nonnegative Lipschitz functions then gives

$$D_y^+ [(|B| + g_N |A|) n^{-\sigma_N}] \leq n^{-\sigma_N} \left(g_N(g|A| + |A'|) - \frac{L}{2} (|B| + g_N |A|) \right).$$

Substituting the identities and factoring out $b_t(n) > 0$ yields the display. $\square$

Lemma 4.8 (Cell decomposition and q -freezing). (i) For $2 \leq n \leq DN$ put $G_0 = \gcd(n, D)$ and list the divisors of G_0 as $1 = d_0 < d_1 < \dots < d_s$ (so $s+1 = \tau(G_0)$), with $d_{-1} = 0$. On the ratio sector $d_{j-1} < n/N \leq d_j$ ($0 \leq j \leq s$) the active set is exactly $\mathcal{D}_{N,n} = \{d_j, \dots, d_s\}$, and for $n/N > d_s$ it is empty (the summand vanishes identically). The number of pairs (mask, sector) with nonempty active set is $\sum_{G_0|D} \tau(G_0) = 3^{|\mathcal{P}|}$.

(ii) With $c = |C(y)|$ and $K = |LC(y) + C'(y)| - \frac{L}{2}c$, so that $G h_+ = U(g) := e^{y(L+g)} [K + gc]_+$, the function $\tilde{g} \mapsto U(\tilde{g})$ is nondecreasing.

(iii) On the sector of (i), within the range $N_{\min} \leq N \leq N_{\max}$ of the prime set, $q_N \geq \max(N_{\min}^2, e^{2L}/d_j^2) - \frac{t_0}{16}$; since g is decreasing in q_N , replacing g by its value at this smallest feasible q_N can only increase $U(g)$.

Proof. (i) $d \in \mathcal{D}_{N,n}$ iff $d \mid G_0$ and $n \leq dN$, i.e. $d \geq n/N$; on the stated sector these are exactly $d_j, \dots, d_s$. Summing $\tau(G_0)$ over the divisors G_0 of the squarefree D gives $\prod_{p \in \mathcal{P}} (\tau(1) + \tau(p)) = 3^{|\mathcal{P}|}$. (ii) Where $K + \tilde{g}c > 0$, $U'(\tilde{g}) = e^{y(L+\tilde{g})} (y(K + \tilde{g}c) + c) \geq 0$; where $K + \tilde{g}c \leq 0$, $U = 0$; and U is continuous. (iii) $n \leq d_j N$ gives $N \geq e^L/d_j$, and $N \geq N_{\min}$; q_N is increasing in N and g decreasing in q_N . $\square$

By Lemma 4.8, for each of the four prime sets the quantity $G h_+$ is bounded by an expression depending only on the cell, on L and on y ; the following finite family of inequalities is therefore checkable by a finite computation.

Proposition 4.9 (Cell inequalities). For each of the four prime sets of (25), every N in the corresponding range, every $2 \leq n \leq DN$ with $\mathcal{D}_{N,n} \neq \emptyset$, and every $y \in [y_0, \sqrt{1-2t_0}]$, in the notation of Lemma 4.7,

$$G h_+ \leq \frac{L}{2} |C_0|.$$

Proof. The [height-transfer program](#) performs the finite calculation that follows. For each prime set it enumerates the $3^{|\mathcal{P}|}$ (mask, ratio-sector) cells of Lemma 4.8(i): for each divisor mask $G_0 \mid D$ and each divisor d_j of G_0 , the active set $\{d_j, \dots, d_s\}$ with the sector constraint $d_{j-1} < n/N \leq d_j$. The (L, y) -domain of the cell, namely

$$L \in \begin{cases} [\log 2, \log(N_{\max} d_0)], & j = 0, \\ [\max(\log 2, \log(N_{\min} d_{j-1})), \log(N_{\max} d_j)], & j \geq 1, \end{cases}$$

and $y \in [y_0, \sqrt{1 - 2t_0}]$, is covered by closed rectangles with outward-padded rational endpoints. Ball arithmetic proves that the exact logarithmic boundaries, y_0 , and $\sqrt{1 - 2t_0}$ lie in this cover. On each rectangle the program replaces q_N by the smallest feasible value $\max(N_{\min}^2, e^{2L}/d_j^2) - \frac{t_0}{16}$ (splitting the L -range at the transition point with overlapping rectangles; conservative by Lemma 4.8(ii)–(iii)), and forms outward enclosures of C_0 , $C(y)$, and $C'(y)$ from the exact identity $r_d(L) = e^{\frac{t_0}{4}(\sum_{p \mid d} \log^2 p + \ell_d^2)} e^{-\frac{t_0}{2} L \ell_d}$, and establishes on each leaf either $h \leq 0$ or the strict inequality $G h_+ < \frac{L}{2} |C_0|$. A rectangle is bisected until one of these two alternatives follows.

The calculation was repeated at 180- and 256-bit precision. At both precisions every rectangle is decided, and the largest ratio satisfies

$$\sup \frac{G h_+}{\frac{L}{2} |C_0|} \leq 0.99999860767275095 < 0.9999988 < 1,$$

the closing comparison being made against the fixed threshold 0.9999988. The worst rectangle lies in the five-prime range at $L \approx 1.042$ – 1.054 , $y \approx 0.1997$ – 0.2022 . Since the closed rectangles cover every cell and the upper ratio is strictly below 1, the required inequality holds throughout the stated domain. $\square$

Proof of Proposition 4.4. Statement (a). For each n with nonempty active set, Lemma 4.7 and Proposition 4.9 give $D_y^+ \leq b_t(n) n^{-\sigma_N} (G h - \frac{L}{2} |C_0|) \leq 0$ on $[y_0, \sqrt{1 - 2t_0}]$ (if $h \leq 0$ both terms are nonpositive; otherwise apply the proposition). Summands with empty active set vanish identically, and the standalone term $g_N(y) = e^{yg}$ is decreasing. Each summand is continuous, and a continuous function with nonpositive upper right Dini derivative on an interval is nonincreasing (if $f(y_2) > f(y_1)$, put $\varepsilon = \frac{f(y_2) - f(y_1)}{2(y_2 - y_1)}$ and let y^* be the supremum of the set where $f(y) \leq f(y_1) + \varepsilon(y - y_1)$; then $y^* < y_2$ and $D^+ f(y^*) \geq \varepsilon > 0$). Hence $\mathcal{F}_{N, \mathcal{P}}$ is nonincreasing.

Conclusion. By Proposition 4.3, $1 - \mathcal{F}_{N, \mathcal{P}}(y_0) = M_N(y_0) L_N(y_0) + M_N(y_0) g_N(y_0) C_N(y_0) \geq M_N(y_0) T_{\min} > 0$. By (a) the numerator $1 - \mathcal{F}_{N, \mathcal{P}}(y)$ is nondecreasing, hence positive on the whole interval; by Lemma 4.6(b) the denominator $M_N(y)$ is positive and nonincreasing; so the quotient $(1 - \mathcal{F}_{N, \mathcal{P}}(y))/M_N(y)$ is nondecreasing. Subtracting the nonnegative nonincreasing correction $g_N(y) C_N(y)$ (Lemma 4.6(c)) preserves this, and $L_N(y) \geq L_N(y_0)$ follows. $\square$

4.5 Error bounds and the finite-range conclusion

The remaining ingredient is a uniform bound on the errors of Theorem 1.3 over the finite region.

Proposition 4.10 (Uniform error bound on the finite region). *For all $x \geq x_*$ and $y_0 \leq y \leq \sqrt{1-2t_0}$, the error terms of Theorem 1.3 at $t = t_0$, bounded using equation (23) of [8] for $e_A + e_B$ and the conservative corollary (19) for $e_{C,0}$, satisfy*

$$e_A + e_B \leq 2.06 \times 10^{-12}, \quad e_{C,0} \leq 0.000000233492848188649183,$$

and

$$e_A + e_B + e_{C,0} \leq E_{\max} := 0.000000233494905212337849.$$

(The third bound is established directly and is sharper than the sum of the two displayed bounds, which are rounded upward for display.)

Proof. The analytic reduction is stated first at the exact time t_0 . The [finite-error program](#) evaluates the same majorants on the slightly larger rational time interval

$$I_t = \left[\frac{129}{800}, \frac{161250001}{10^9} \right],$$

so its result is a conservative strengthening of the calculation needed here. All integers and rational constants are constructed exactly; square roots, logarithms, exponentials, powers, and π are evaluated with outward-rounded FLINT/Arb balls. Whenever an upper endpoint is reused in a monotone majorant, the program first converts that endpoint to an exact dyadic number, so no midpoint approximation enters the proof.

Set $N_0 := 690988$. Evaluation at the smaller window edge

$$x_0 := x_{N_0} = 4\pi \left(N_0^2 - \frac{t_0}{16} \right) < x_*$$

(the strict inequality is (21)) dominates the whole region $x \geq x_*$, $y_0 \leq y \leq \sqrt{1-2t_0}$, $t = t_0$. The x -dependent factors used below are bounded at x_0 ; the dependence on the changing window index is handled separately in part (iii). Write $\ell_0 = \log \frac{x_0}{4\pi} = \log(N_0^2 - \frac{t_0}{16})$,

$$\delta_1 = \frac{t_0}{4} \log \left(1 - \frac{t_0}{16N_0^2} \right)^{-1} + \frac{t_0}{2x_0^2}, \quad \sigma_1 = \frac{1+y_0}{2} + \frac{t_0}{2} \log N_0 - \delta_1 = \frac{1+y_0}{2} + \frac{t_0}{4} \ell_0 - \frac{t_0}{2x_0^2}.$$

More generally, for $N \geq N_0$ put

$$\delta_N := \frac{t_0}{4} \log \left(1 - \frac{t_0}{16N^2} \right)^{-1} + \frac{t_0}{2x_N^2}, \quad \tilde{\sigma}_N := \frac{1+y_0}{2} + \frac{t_0}{2} \log N - \delta_N.$$

Thus $\tilde{\sigma}_{N_0} = \sigma_1$. Both summands of δ_N decrease with N , so $\delta_N \leq \delta_1$ and $\tilde{\sigma}_N$ grows with $\log N$ at rate at least $t_0/2$. The exponent estimate used in part (i) below also gives $\operatorname{Re} s_* \geq \tilde{\sigma}_N$ throughout window W_N .

(i) *The exponent.* In (17) the positive part is at most 1: since $y_0 < 1$ and $8/x^2 \leq 8/x_0^2 < 3y_0 < 3$, $\frac{4y(1+y)}{x^2} \leq \frac{8y}{x^2} < 3y$ for all $y \geq y_0$, and $\operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t_0}{4} \log \frac{x}{4\pi} - \frac{t_0}{2x^2} \geq \sigma_1$ throughout the region.

(ii) *The bound (19) for $e_{C,0}$.* Each factor is monotone in the favorable direction: the prefactor $(\frac{x}{4\pi})^{-(1+y)/4}$ decreases in x and y , while $-\frac{t_0}{16} \log^2 \frac{x}{4\pi}$ decreases in x and is independent of y (here $\ell_0 > 2$). The factor $3^y + 3^{-y}$ increases in y ; $N - 0.125$ increases with the window; and the final quotient decreases in x because $3 \frac{d}{dx} |\log \frac{x}{4\pi} + \frac{i\pi}{2}| \leq \frac{3}{x}$ while the numerator exceeds $10.50 > 3$. Hence (19) evaluated at $x = x_0$ and $N = N_0$, with $y = y_0$ in the decreasing factors

and $y = \sqrt{1 - 2t_0}$ in $3^y + 3^{-y}$, bounds $e_{C,0}$ on the whole region; this evaluation is the program's quantity `error_c0`.

(iii) *The bound (23) for $e_A + e_B$.* Since $4\pi n^2 \leq 4\pi N^2 \leq x(1 - \frac{t_0}{16N^2})^{-1}$ for every $n \leq N$, the bounds (16) and (18) give, uniformly on the region,

$$1 + |\gamma|N^{|\kappa|}n^y \leq 1 + e^{0.02} \left(1 - \frac{t_0}{16N_0^2}\right)^{-1/2} \exp\left(\frac{t_0 \log N_0}{2(x_0 - 6)}\right)$$

(the quantity $\log N/(x_N - 6)$ decreases in N). To control the logarithm in the last factor, put $r = x/(4\pi n^2)$. Since $n \leq N$ and $x \geq 4\pi(N^2 - t_0/16)$,

$$r \geq 1 - \frac{t_0}{16N^2}, \quad r \leq \frac{x}{4\pi}.$$

Moreover,

$$\frac{x}{4\pi} \left(1 - \frac{t_0}{16N^2}\right) \geq \frac{(N^2 - t_0/16)^2}{N^2} \geq 1,$$

where the last inequality follows from $N^2 - t_0/16 \geq N$. Thus, whether r is below or above 1, one has $|\log r| \leq \log(x/(4\pi))$. Consequently $\log^2 \frac{x}{4\pi n^2} \leq \log^2 \frac{x}{4\pi}$, and the monotonicity of $\log^2(\frac{x}{4\pi})/(x - 6.66)$ in x (using $\ell_0 > 2$) bounds the last factor of (23) by $\exp(\frac{(t_0^2/16)\ell_0^2 + 0.626}{x_0 - 6.66}) - 1$. For the middle factor, $\sum_{n=1}^N b_{t_0}(n) n^{-\text{Re } s^*} \leq 1 + \sum_{n=2}^{m_0} b_{t_0}(n) n^{-\sigma_1} + T_N$ with $m_0 = 2000$, where $T_N := \sum_{m_0 < n \leq N} b_{t_0}(n) n^{-\tilde{\sigma}_N}$. The summand $\exp(\frac{t_0}{4} \log^2 n - \tilde{\sigma}_N \log n)$ decreases in n on $n \leq N$ because $\frac{1+y_0}{2} - \delta_1 > 0$, so $T_N \leq \int_{\log m_0}^{\log N} e^{\psi(L)} dL$ with $\psi(L) = (1 - \tilde{\sigma}_N)L + \frac{t_0}{4}L^2$; since ψ is convex, $T_N \leq (\log N - \log m_0) \max(e^{\psi(\log m_0)}, e^{\psi(\log N)})$. Finally, the two rate inequalities

$$\frac{t_0}{2} \log m_0 > \left(\log \frac{N_0}{m_0}\right)^{-1}, \quad \frac{t_0}{2} \log N_0 - \frac{1 - y_0}{2} - \delta_1 > \left(\log \frac{N_0}{m_0}\right)^{-1},$$

imply that each of the two endpoint products $(\log N - \log m_0)e^{\psi(\log m_0)}$ and $(\log N - \log m_0)e^{\psi(\log N)}$ is nonincreasing in N for $N \geq N_0$ (using the growth of $\tilde{\sigma}_N$ just noted and $\delta_N \leq \delta_1$), so their values at $N = N_0$ dominate every window; the program proves both rate inequalities. The product of the three bounds is its quantity `error_ab`.

The program implements the five majorants just described, proves all twelve domain and sign inequalities used in the monotonicity argument, and bounds the head sum for $2 \leq n \leq 2000$ term by term over the whole time interval I_t . It then compares the resulting balls with exact rational right-hand sides. Put $\mathcal{E} = e_A + e_B + e_{C,0}$. Runs at 256 and 512 bits give

$$\begin{aligned} e_A + e_B &< 2.05702368866622 \times 10^{-12}, \\ e_{C,0} &< \frac{233492848188649183}{10^{24}}, \\ \mathcal{E} &< \frac{233494905212337849}{10^{24}} = E_{\max}, \\ T_{\min} - \mathcal{E} &> 5.5787109478766215 \times 10^{-7}. \end{aligned}$$

Both runs also prove directly $\mathcal{E} < 234 \times 10^{-9}$ and $T_{\min} - \mathcal{E} > 557 \times 10^{-9} > 0$. Thus both precisions prove the same strict rational inequalities, and hence all three bounds in the statement. $\square$

We can now state the conclusion of the finite range.

Proposition 4.11 (Nonvanishing at time t_0 , finite range). $H_{t_0}(x + iy) \neq 0$ for all $x_* \leq x < x_{3840001}$ and $y_0 \leq y \leq \sqrt{1 - 2t_0}$.

Proof. Let x be in the stated range and let N be its window index, so that $690988 \leq N \leq 3840000$ by (21), and $x \in W_N$. For y in the stated range, Lemma 4.2 (applied with the height y , the prime set $\mathcal{P}(N)$, and the frozen constants of Lemma 4.1), together with Propositions 4.3 and 4.4, gives

$$|f_{t_0}(x + iy)| \geq L_N(y) \geq L_N(y_0) \geq T_{\min}.$$

Combining with Proposition 4.10,

$$|f_{t_0}(x + iy)| - (e_A + e_B + e_{C,0}) \geq T_{\min} - E_{\max} \geq 0.000000557871094787 > 0. \quad (26)$$

The point (x, y, t_0) lies in the region (10), so [8, Corollary 1.4] (the criterion for non-vanishing quoted in Section 2.2) yields $H_{t_0}(x + iy) \neq 0$. $\square$

5 Hypothesis (ii), infinite tail

This section proves nonvanishing of H_t for *all* windows $N \geq N_* = 3840000$, all $t \in I_t = [\frac{129}{800}, \frac{161250001}{10^9}]$ (a closed interval whose left endpoint is exactly t_0), and all $y \in [y_0, \sqrt{1 - 2t}]$, by a single interval computation at the cutoff N_* combined with monotonicity arguments in N and y . Since $t_0 \in I_t$, this in particular covers the case $t = t_0$ needed for hypothesis (ii); the last window $W_{3840000}$ of the finite range overlaps the tail's starting abscissa $x_{3840000}$, so the two ranges jointly cover all $x \geq x_*$.

5.1 Setup: mollifier and frozen exponents

Fix the sparse mollifier support $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 10, 11, 13, 14\}$ and the exact rational vector

$$\begin{array}{c|cccccccccccc} d & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 10 & 11 & 13 & 14 \\ \hline \lambda_d & 1 & -\frac{1021}{1000} & -\frac{1054}{1000} & -\frac{9}{200} & -\frac{1119}{1000} & \frac{1001}{1000} & -\frac{1043}{1000} & \frac{128}{125} & -\frac{161}{200} & -\frac{447}{500} & \frac{456373}{500000} \end{array}$$

($\lambda_1 = 1$; $\sum_d |\lambda_d| = 9.918746$ exactly), and set $M_\lambda(z) = \sum_{d \in \mathcal{S}} \lambda_d d^{-z}$. (Unlike the Euler-product mollifier of Section 4.2, this vector was chosen by numerical optimization; its only structural requirement is $\lambda_1 = 1$ and real entries.) Fix also the head lengths $M = 153814$ and $M_{\text{err}} = 3000$.

For $x \geq X_N(t) := 4\pi(N^2 - \frac{t}{16})$ with window index $N \geq N_*$, set

$$\delta(N, t) = \frac{t}{4} \left[-\log\left(1 - \frac{t}{16N^2}\right) \right] + \frac{t}{2X_N(t)^2}, \quad k(N, t) = \frac{t}{2(X_N(t) - 6)}.$$

Both functions decrease in N for fixed t . Define the frozen constants exactly by the compact-interval maxima

$$\widehat{\delta} := \max_{t \in I_t} \delta(N_*, t), \quad \widehat{k} := \max_{t \in I_t} k(N_*, t).$$

They also increase in t on I_t , so these maxima equal their values at $t = 161250001/10^9$. The frozen exponents are then

$$\sigma_1 = \frac{1+y}{2} + \frac{t}{2} \log N - \widehat{\delta}, \quad \sigma_2 = \frac{1-y}{2} + \frac{t}{2} \log N - \widehat{\delta} - \widehat{k}, \quad G = e^{0.02y} \left(N^2 - \frac{t}{16} \right)^{-y/2}.$$

Lemma 5.1 (Frozen exponents). *Let $N \geq N_*$, $t \in I_t$, $0 < y \leq 1$, and let $x \in [X_N(t), X_{N+1}(t))$, so that x has window index N . Then*

$$|\gamma| \leq G, \quad \operatorname{Re} s_* \geq \sigma_1, \quad |\kappa| \leq \widehat{k},$$

and consequently each term of the second sum in (12) satisfies

$$|n^y b_t(n) n^{-\overline{s_*} - \kappa}| \leq b_t(n) n^{-\sigma_2} \quad (1 \leq n \leq N).$$

Proof. Since $x \geq X_N(t)$ we have $x/4\pi \geq N^2 - t/16$, and the three factors $(x/4\pi)^{-y/2}$, x^{-2} , $(x-6)^{-1}$ appearing in (16)–(18) are decreasing in x , hence maximal at the window's left edge. For γ this gives $|\gamma| \leq e^{0.02y}(N^2 - t/16)^{-y/2} = G$ directly. For κ , using $y \leq 1$ and that $X_N(t)$ increases with N ,

$$|\kappa| \leq \frac{ty}{2(x-6)} \leq \frac{t}{2(X_N(t)-6)} = k(N, t) \leq k(N_*, t) \leq \widehat{k}.$$

For $\operatorname{Re} s_*$: by the exact identity

$$\frac{t}{4} \log \left(N^2 - \frac{t}{16} \right) = \frac{t}{2} \log N - \frac{t}{4} \left[-\log \left(1 - \frac{t}{16N^2} \right) \right],$$

the logarithmic term of (17) is at least $\frac{t}{2} \log N$ minus the first summand of $\delta(N, t)$. The subtracted positive-part term of (17) is at most $t/(2x^2) \leq t/(2X_N(t)^2)$, the second summand of $\delta(N, t)$: indeed, on the domain $0 \leq y \leq 1$, $x \geq 200$ one has $1 - 3y + 4y(1+y)/x^2 \leq 1$ (for $y > 0$ this is $4(1+y) \leq 3x^2$, and at $y = 0$ the left side equals 1), so the positive part is at most 1. Both summands of $\delta(N, t)$ decrease in N (in the first, $t/16N^2$ decreases and $u \mapsto -\log(1-u)$ is increasing; in the second, $X_N(t)$ increases), so

$$\operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t}{2} \log N - \delta(N, t) \geq \frac{1+y}{2} + \frac{t}{2} \log N - \widehat{\delta} = \sigma_1.$$

Finally, for $1 \leq n \leq N$,

$$|n^y b_t(n) n^{-\overline{s_*} - \kappa}| = b_t(n) n^{y - \operatorname{Re} s_* - \operatorname{Re} \kappa} \leq b_t(n) n^{y - \sigma_1 + \widehat{k}} = b_t(n) n^{-\sigma_2},$$

since $-\operatorname{Re} \kappa \leq |\kappa| \leq \widehat{k}$ and $\sigma_2 = \sigma_1 - y - \widehat{k}$. □

5.2 The endpoint-cap lemma

The analytic engine for tail sums is:

Lemma 5.2 (Endpoint cap). *For $1 \leq a < c$ set $E_{t,\sigma}(u) = \exp((1-\sigma) \log u + \frac{t}{4} \log^2 u)$ and*

$$\operatorname{Cap}_t(a, c; \sigma) = \max(E_{t,\sigma}(a), E_{t,\sigma}(c)) \log \frac{c}{a}.$$

If $\sigma > \frac{t}{2} \log c$, then for all integers $a \leq J < K \leq c$,

$$\sum_{J < n \leq K} b_t(n) n^{-\sigma} \leq \operatorname{Cap}_t(a, c; \sigma).$$

Proof. The logarithmic derivative of $b_t(u)u^{-\sigma}$ in $\log u$ is $\frac{t}{2} \log u - \sigma < 0$ on $[a, c]$, so the summand decreases and the sum is at most $\int_a^c b_t(u)u^{-\sigma} du$. Substituting $v = \log u$ makes the integrand $\exp((1-\sigma)v + \frac{t}{4}v^2)$, which is log-convex in v , hence maximized at an endpoint of $[\log a, \log c]$; multiplying by the length $\log(c/a)$ of the v -interval gives the bound. □

5.3 Exact convolution and the contraction D

Define, for $N \geq N_*$ and (x, y, t) with x in the window of index N ,

$$P = \sum_{m=2}^M |c_m(t)| m^{-\sigma_1}, \quad c_m(t) = \sum_{\substack{d \in \mathcal{S} \\ d|m}} \lambda_d b_t(m/d), \quad M_{\max} = M_{\max}(N, t, y) := \sum_{d \in \mathcal{S}} |\lambda_d| d^{-\sigma_1},$$

$$\text{TR} = \sum_{d \in \mathcal{S}} |\lambda_d| d^{-\sigma_1} \text{Cap}_t(\lfloor M/d \rfloor, N; \sigma_1), \quad \text{AB} = G\left(\sum_{k \leq M} b_t(k) k^{-\sigma_2} + \text{Cap}_t(M, N; \sigma_2)\right),$$

together with the nonnegative “overshoot” term

$$\text{OV} = \sum_{\substack{d \in \mathcal{S} \\ d > 1}} |\lambda_d| d^{-\sigma_1} \text{Cap}_t\left(\frac{N}{d+1}, N; \sigma_1\right)$$

(included for compatibility with the program of Proposition 5.10; see the remark after the proof), and set

$$D = D(N, t, y) := P + \text{TR} + \text{OV} + M_{\max} \text{AB}. \quad (27)$$

Lemma 5.3 (Mollified contraction bound). *Assume the cap-validity conditions*

$$\widehat{\delta} < \frac{1+y}{2}, \quad \widehat{\delta} + \widehat{k} < \frac{1-y}{2} \quad (28)$$

(equivalently $\sigma_1 > \frac{t}{2} \log N$ and $\sigma_2 > \frac{t}{2} \log N$: the differences $\sigma_i - \frac{t}{2} \log N$ are independent of N). Then for every $N \geq N_*$, $t \in I_t$, $0 < y \leq 1$ and x in the window of index N ,

$$|M_\lambda(s_*) f_t(x + iy) - 1| \leq D.$$

Consequently, if $D < 1$ then

$$|f_t(x + iy)| \geq \frac{1-D}{M_{\max}}. \quad (29)$$

Proof. Write the two sums in (12) as F_B and F_A ($f_t = F_B + \gamma F_A$). Expanding $M_\lambda(s_*) F_B = \sum_{d \in \mathcal{S}} \sum_{k \leq N} \lambda_d b_t(k) (dk)^{-s_*}$ and partitioning the index pairs (d, k) by $dk \leq M$ versus $dk > M$ — a disjoint and exhaustive partition — gives the *exact* identity

$$M_\lambda(s_*) F_B - 1 = \sum_{m=2}^M c_m(t) m^{-s_*} + \sum_{d \in \mathcal{S}} \lambda_d d^{-s_*} \sum_{\lfloor M/d \rfloor < k \leq N} b_t(k) k^{-s_*}. \quad (30)$$

Here the first term groups the pairs with $dk \leq M$ by $m = dk$: since $m \leq M < N_* \leq N$, the constraint $k = m/d \leq N$ is automatic, so the coefficient of m^{-s_*} is exactly $c_m(t)$, and the $m = 1$ term $c_1 = \lambda_1 b_t(1) = 1$ has been moved to the left side. In the second term, for fixed d the condition $dk > M$ for integer k is $k > \lfloor M/d \rfloor$. By $\text{Re } s_* \geq \sigma_1$ (Lemma 5.1), the first term is bounded by P ; and each inner sum of the second term is bounded, by Lemma 5.2 with $(a, c) = (\lfloor M/d \rfloor, N)$, by $\text{Cap}_t(\lfloor M/d \rfloor, N; \sigma_1)$, so the second term is bounded by TR ; the application of Lemma 5.2 is legitimate because its hypothesis $\sigma_1 > \frac{t}{2} \log N$ is the first condition of (28). On the A -side, Lemma 5.1 bounds each term of $|\gamma F_A|$ by $G b_t(k) k^{-\sigma_2}$; splitting at $k = M$ and applying Lemma 5.2 with $(a, c) = (M, N)$ at exponent σ_2 (legitimate by the

second condition of (28)) gives $|\gamma F_A| \leq AB$, while $|M_\lambda(s_*)| \leq \sum_d |\lambda_d| d^{-\sigma_1} = M_{\max}$; hence $|M_\lambda(s_*)\gamma F_A| \leq M_{\max}AB$. Adding the nonnegative OV only weakens the bound, and the first claim follows. For the second claim: if $D < 1$ then $|M_\lambda(s_*)f_t| \geq 1 - D > 0$; in particular $M_\lambda(s_*) \neq 0$, and dividing by $|M_\lambda(s_*)| \leq M_{\max}$ gives $|f_t| \geq (1 - D)/M_{\max}$. $\square$

Remark 5.4. The term OV plays no role in the mathematics — the identity (30) is fully covered by $P + \text{TR}$ — but the program of Proposition 5.10 below includes it in D , so it is included in (27) in order that the numerical bound on D majorizes the same quantity the program computes.

5.4 Reduction to the cutoff

The interval computation of Section 5.6 evaluates D , and the error majorants, only at the cutoff $N = N_*$, over the closed intervals

$$I_{\text{box}} = \left[\frac{1872719}{10^7}, \frac{23409}{125000} \right], \quad I_{\text{ext}} = \left[\frac{1872719}{10^7}, \frac{8231039}{10^7} \right],$$

where I_{box} contains y_0 and the top of I_{ext} is at least $\sqrt{1 - 2t}$ for every $t \in I_t$. The reduction from this single evaluation to all $N \geq N_*$ and the full height range is carried out in Lemmas 5.5–5.9 below. Throughout, the *cutoff conditions* are the following inequalities, all evaluated at $N = N_*$, uniformly for $t \in I_t$ and $y \in I_{\text{ext}}$:

$$\frac{t}{2} \log a (\log N_* - \log a) > 1 \quad \text{and} \quad (\sigma - 1)(\log N_* - \log a) > 1 \quad (31)$$

for each cap $\text{Cap}_t(a, \cdot; \sigma)$ appearing in D and in the error bounds (the left endpoints $a = \lfloor M/d \rfloor$ for $d \in \mathcal{S}$, $a = M$, and $a = M_{\text{err}}$, with $\sigma = \sigma_1$ or σ_2 as appropriate), together with the A -side slope condition

$$0.02 - \frac{1}{2} \log(N_*^2 - \frac{t}{16}) + \frac{1}{2} \log N_* < 0. \quad (32)$$

Lemma 5.5 (Decay of fixed-endpoint caps). *Let $t > 0$, let $1 < a < N_*$, let $\sigma(q) = \sigma_0 + \frac{t}{2}q$ with $\sigma_0 \in \mathbb{R}$, and write $q_* = \log N_*$. Assume the two conditions (31) for this pair, that is, $\frac{t}{2} \log a (q_* - \log a) > 1$ and $(\sigma(q_*) - 1)(q_* - \log a) > 1$. Then both conditions persist for all $q \geq q_*$, one has $\sigma(q) > 1$ for all $q \geq q_*$, and*

$$q \mapsto \text{Cap}_t(a, e^q; \sigma(q))$$

is positive and decreasing on $[q_, \infty)$.*

Proof. By definition $\text{Cap}_t(a, e^q; \sigma(q)) = \max(F_1(q), F_2(q))$, where

$$F_1(q) = E_{t, \sigma(q)}(a) (q - \log a), \quad F_2(q) = E_{t, \sigma(q)}(e^q) (q - \log a).$$

Since $\log E_{t, \sigma}(u) = (1 - \sigma) \log u + \frac{t}{4} \log^2 u$ and $\sigma'(q) = \frac{t}{2}$,

$$\frac{d}{dq} \log F_1(q) = -\frac{t}{2} \log a + \frac{1}{q - \log a},$$

which is negative exactly when $\frac{t}{2} \log a (q - \log a) > 1$; both factors on the left of this condition are positive and nondecreasing in q , so the condition, true at q_* , persists for all $q \geq q_*$. Similarly,

$$\frac{d}{dq} \log F_2(q) = \left[(1 - \sigma(q)) - \frac{t}{2} q \right] + \frac{t}{2} q + \frac{1}{q - \log a} = 1 - \sigma(q) + \frac{1}{q - \log a},$$

which is negative exactly when $(\sigma(q) - 1)(q - \log a) > 1$. At q_* this condition forces $\sigma(q_*) > 1$; then both factors $\sigma(q) - 1$ and $q - \log a$ are positive and increasing in q , so the condition persists, and $\sigma(q) > 1$ throughout. Hence F_1 and F_2 are positive and decreasing on $[q_*, \infty)$, and so is their maximum. $\square$

Lemma 5.6 (Componentwise decay in N). *Fix $t \in I_t$ and $y \in (0, 1]$, and assume the cutoff conditions (31). Then each of the following is nonincreasing in N on $[N_*, \infty)$ (throughout, $q = \log N$ and the frozen exponents $\sigma_i = \sigma_i(q)$ are increasing in q with slope $\frac{t}{2}$):*

- (i) $m^{-\sigma_i(q)}$ for every fixed integer $m \geq 2$ and $i \in \{1, 2\}$; hence the head P , the mollifier bound $M_{\max}$, and the A -side head $\sum_{k \leq M} b_t(k) k^{-\sigma_2}$ (whose $k = 1$ term is constant) are nonincreasing;
- (ii) $G = e^{0.02y} (e^{2q} - \frac{t}{16})^{-y/2}$;
- (iii) the moving-endpoint term OV .

Proof. (i) is immediate: $\sigma_i(q)$ is increasing and $\log m > 0$, while the coefficients $|c_m(t)|$, $|\lambda_d|$ and $b_t(k)$ do not depend on N . For (ii),

$$\frac{d}{dq} \log G = -\frac{y}{2} \cdot \frac{2e^{2q}}{e^{2q} - t/16} = -y \frac{N^2}{N^2 - t/16} < 0.$$

For (iii), each summand of OV is $|\lambda_d| d^{-\sigma_1(q)} \text{Cap}_t(e^q/(d+1), e^q; \sigma_1(q))$, a product of the nonincreasing factor $d^{-\sigma_1(q)}$ ($d \geq 2$) with a cap whose endpoints both scale with N and whose length $\log(d+1)$ is constant. For $r \in \{1, d+1\}$,

$$\frac{d}{dq} \left[(1 - \sigma(q))(q - \log r) + \frac{t}{4}(q - \log r)^2 \right] = (1 - \sigma(q)) - \frac{t}{2}(q - \log r) + \frac{t}{2}(q - \log r) = 1 - \sigma(q) < 0,$$

using $\sigma_1(q) > 1$ from Lemma 5.5. Thus both endpoint expressions are positive and decreasing, so their maximum and the summand are decreasing as well. $\square$

Lemma 5.7 (Decay in y at the cutoff). *Fix $t \in I_t$ and assume the A -side slope condition (32). Then $y \mapsto D(N_*, t, y)$ and $y \mapsto M_{\max}(N_*, t, y)$ are nonincreasing on $(0, 1]$.*

Proof. Recall $\partial \sigma_1 / \partial y = \frac{1}{2}$ and $\partial \sigma_2 / \partial y = -\frac{1}{2}$. On the B -side, every y -dependent factor has the form $u^{-\sigma_1(y)}$ or $E_{t, \sigma_1(y)}(u)$ with $u > 1$ (the head terms $m \geq 2$, the mollifier terms $d \geq 2$, and the cap ingredients at $u = \lfloor M/d \rfloor$, $u = N_*$ or $u = N_*/r$, all exceeding 1), and satisfies $\partial_y \log(\cdot) = -\frac{1}{2} \log u < 0$; the remaining factors ($|c_m(t)|$, $|\lambda_d|$, b_t , cap lengths) are independent of y , and maxima of nonincreasing functions are nonincreasing. Hence P , TR , OV and $M_{\max}$ are nonincreasing in y . On the A -side, $\log G$ is linear in y with slope $0.02 - \frac{1}{2} \log(N_*^2 - \frac{t}{16})$, so each head term and each cap ingredient of AB , with $u \leq N_*$, satisfies

$$\partial_y \log(G b_t(u) u^{-\sigma_2(y)}) \text{ resp. } \partial_y \log(G E_{t, \sigma_2(y)}(u)) = 0.02 - \frac{1}{2} \log(N_*^2 - \frac{t}{16}) + \frac{1}{2} \log u,$$

which is at most the left side of (32), hence negative. So AB is nonincreasing in y , and therefore so are $M_{\max}AB$ and D . $\square$

Lemma 5.8 (Reduction to the cutoff). *Assume the cutoff conditions (31)–(32) hold. Then for every $N \geq N_*$, every $t \in I_t$ and every $y \in [y_0, \sqrt{1-2t}]$,*

$$D(N, t, y) \leq \sup_{y' \in I_{\text{box}}} D(N_*, t, y'), \quad M_{\max}(N, t, y) \leq \sup_{y' \in I_{\text{box}}} M_{\max}(N_*, t, y').$$

Proof. Fix t and y . Every constituent of $D = P + \text{TR} + \text{OV} + M_{\max} \text{AB}$ is a sum of products of nonnegative factors, each nonincreasing in N on $[N_*, \infty)$: the heads, $M_{\max}$ and G by Lemma 5.6(i)–(ii); the fixed-endpoint caps in TR and AB by Lemma 5.5, applied with $a = \lfloor M/d \rfloor$ (at σ_1) and $a = M$ (at σ_2), whose conditions (31) hold by hypothesis; and OV by Lemma 5.6(iii). Sums and products of nonnegative nonincreasing functions are nonincreasing, so $D(N, t, y) \leq D(N_*, t, y)$ and $M_{\max}(N, t, y) \leq M_{\max}(N_*, t, y)$. If $y \in I_{\text{box}}$ the claims follow at once. Otherwise $y > \max I_{\text{box}}$, since $y \geq y_0 \geq \min I_{\text{box}}$; as $y \leq \sqrt{1-2t} < 1$, Lemma 5.7 applies on $[\max I_{\text{box}}, y]$ and gives $D(N_*, t, y) \leq D(N_*, t, \max I_{\text{box}}) \leq \sup_{y' \in I_{\text{box}}} D(N_*, t, y')$, and the same for $M_{\max}$. $\square$

5.5 Tail error terms

The error bounds use the exact definitions (71)–(72) of [8] rather than the coarser display (23): (23) applies the worst-case factor $N^{|\kappa|}$ to every term, whereas (71) carries the per-term factor $n^{-\Re \kappa}$, which Lemma 5.1 bounds by $\widehat{n^{\widehat{\kappa}}}$. Define, for $N \geq N_*$, $t \in I_t$ and $0 < y \leq 1$,

$$L_N(t) = \log\left(N^2 - \frac{t}{16}\right), \quad \Delta(N, t) = \frac{\frac{t^2}{16} L_N(t)^2 + 0.626}{X_N(t) - 6.66},$$

and the two majorants

$$\begin{aligned} E_{AB}(N, t, y) &= (e^{\Delta(N, t)} - 1)(S_1(N, t, y) + G S_2(N, t, y)), \\ S_i(N, t, y) &= \sum_{n=1}^{M_{\text{err}}} b_t(n) n^{-\sigma_i} + \text{Cap}_t(M_{\text{err}}, N; \sigma_i) \quad (i = 1, 2), \\ E_C(N, t, y) &= \left(N^2 - \frac{t}{16}\right)^{-(1+y)/4} \\ &\quad \times \exp\left(-\frac{t}{16} L_N(t)^2 + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{3\sqrt{L_N(t)^2 + \pi^2/4} + 10.50}{X_N(t) - 12}\right). \end{aligned}$$

Lemma 5.9 (Error majorants). *Assume the cap-validity conditions (28) and the conditions (31) for the caps with left endpoint $a = M_{\text{err}}$. Then for every $N \geq N_*$, $t \in I_t$, $y \in [y_0, \sqrt{1-2t}]$ and x in the window of index N ,*

$$e_A + e_B \leq E_{AB}(N, t, y) \leq E_{AB}(N_*, t, y), \quad e_{C,0} \leq E_C(N, t, y) \leq E_C(N_*, t, y).$$

Proof. The per-term factor. The quantity $\varepsilon_{t,n}$ of [8, (44)] is bounded, by the display in the proof of [8, Proposition 6.6(iv),(v)],

$$\varepsilon_{t,n}\left(\frac{1 \pm y + ix}{2}\right) \leq \exp\left(\frac{\frac{t^2}{32} \log^2 \frac{x}{4\pi n^2} + 0.313}{\frac{x}{2} - 3.33}\right) - 1 = \exp\left(\frac{\frac{t^2}{16} \log^2 \frac{x}{4\pi n^2} + 0.626}{x - 6.66}\right) - 1,$$

the imaginary part of $\frac{1 \pm y + ix}{2}$ being $\frac{x}{2}$. For $1 \leq n \leq N$ and x in the window of index N one has $\frac{x}{4\pi n^2} \leq \frac{x}{4\pi}$ and $\frac{x}{4\pi n^2} \geq \frac{X_N(t)}{4\pi N^2} = 1 - \frac{t}{16N^2}$, so

$$\left| \log \frac{x}{4\pi n^2} \right| \leq \max \left(\log \frac{x}{4\pi}, -\log \left(1 - \frac{t}{16N^2} \right) \right) = \log \frac{x}{4\pi},$$

the last step because $\frac{x}{4\pi} \left(1 - \frac{t}{16N^2} \right) \geq \frac{(N^2 - t/16)^2}{N^2} \geq 1$ (indeed $N^2 - \frac{t}{16} \geq N$ as $N \geq N_*$). Hence the per-term exponent is at most $\varphi(x) = (\frac{t^2}{16} \log^2 \frac{x}{4\pi} + 0.626)/(x - 6.66)$. Writing $L = \log \frac{x}{4\pi}$, which exceeds 2 for $x \geq 200$,

$$\begin{aligned} (x - 6.66)^2 \varphi'(x) &= \frac{t^2}{8} L \frac{x - 6.66}{x} - \left(\frac{t^2}{16} L^2 + 0.626 \right) \\ &< \frac{t^2}{8} L - \frac{t^2}{16} L^2 - 0.626 \\ &= -\frac{t^2}{16} L(L - 2) - 0.626 < 0, \end{aligned}$$

so φ is decreasing, and on the window $\varphi(x) \leq \varphi(X_N(t)) = \Delta(N, t)$: the per-term factor is at most $e^{\Delta(N, t)} - 1$.

Assembly of E_{AB} . By (71)–(72) of [8] and Lemma 5.1 ($\operatorname{Re} s_* \geq \sigma_1$, $|\gamma| \leq G$, and $n^{y - \operatorname{Re} s_* - \operatorname{Re} \kappa} \leq n^{-\sigma_2}$),

$$e_B \leq (e^{\Delta(N, t)} - 1) \sum_{n \leq N} b_t(n) n^{-\sigma_1}, \quad e_A \leq G(e^{\Delta(N, t)} - 1) \sum_{n \leq N} b_t(n) n^{-\sigma_2}.$$

Splitting each sum at M_{err} and bounding the range $M_{\text{err}} < n \leq N$ by Lemma 5.2 (legitimate by (28)) gives $e_A + e_B \leq E_{AB}(N, t, y)$.

Bound for $e_{C,0}$. By the corollary (19), $e_{C,0}$ is bounded by the product of $(\frac{x}{4\pi})^{-(1+y)/4}$, $\exp(-\frac{t}{16} \log^2 \frac{x}{4\pi})$, $\exp(1.24(3^y + 3^{-y})/(N - 0.125))$, and $\exp(\psi(x))$ with $\psi(x) = (3\sqrt{L^2 + \pi^2/4} + 10.50)/(x - 12)$. The first two factors are decreasing in x , the third does not depend on x , and ψ is decreasing: with $Q(L) = 3\sqrt{L^2 + \pi^2/4} + 10.50$ one has $\frac{d}{dx} Q = \frac{3L}{\sqrt{L^2 + \pi^2/4}} \cdot \frac{1}{x} < \frac{3}{x}$, whence $(x - 12)^2 \psi'(x) < 3 \frac{x-12}{x} - Q < 3 - 10.50 < 0$. Evaluating all four factors at the left edge $x = X_N(t)$, where $\log \frac{x}{4\pi} = L_N(t)$, gives $e_{C,0} \leq E_C(N, t, y)$.

Decay in N . In E_{AB} : the factor $e^{\Delta(N, t)} - 1$ is nonincreasing because $\Delta(N, t) = \varphi(X_N(t))$ with φ decreasing and $X_N(t)$ increasing in N ; the heads of S_1, S_2 and the factor G are nonincreasing by Lemma 5.6; and the caps $\operatorname{Cap}_t(M_{\text{err}}, N; \sigma_i)$ are nonincreasing by Lemma 5.5 with $a = M_{\text{err}}$, whose conditions (31) hold by hypothesis. In E_C : the factors $(N^2 - t/16)^{-(1+y)/4}$, $e^{-\frac{t}{16} L_N(t)^2}$ and $e^{1.24(3^y + 3^{-y})/(N - 0.125)}$ are visibly nonincreasing in N , and $e^{\psi(X_N(t))}$ is nonincreasing because ψ is decreasing and $X_N(t)$ increases. Products of nonnegative nonincreasing factors are nonincreasing. $\square$

5.6 The cutoff inequalities and the tail theorem

The computational input of the tail argument is the following.

Proposition 5.10 (Cutoff inequalities). *The cap-validity conditions (28) and the cutoff conditions (31)–(32) hold, uniformly for $t \in I_t$ and $y \in [y_0, \sqrt{1 - 2t}]$, and*

$$\sup_{t \in I_t, y \in I_{\text{box}}} D(N_*, t, y) < 0.999721, \quad \sup_{t \in I_t, y \in I_{\text{box}}} M_{\max}(N_*, t, y) < 1.608290,$$

$$\sup_{t \in I_t, y \in I_{\text{ext}}} (E_{AB} + E_C)(N_*, t, y) < 1.1672 \times 10^{-8},$$

with E_{AB}, E_C the error majorants of Lemma 5.9. In addition, for every $N \geq N_*$, $t \in I_t$, $y \in [y_0, \sqrt{1-2t}]$, and $x \in [X_N(t), X_{N+1}(t))$, the inequality

$$\frac{1 - D(N, t, y)}{M_{\max}(N, t, y)} - (e_A(x, y, t) + e_B(x, y, t) + e_{C,0}(x, y, t)) > 0.00017352 > 0.$$

Proof. The `tail program` evaluates the quantities below from the exact rational parameters and the eleven rational mollifier coefficients. It uses outward-rounded FLINT/Arb balls and reports an unresolved comparison if an interval overlaps a required boundary. The calculation was repeated at 256 and 512 bits.

How the suprema are captured. The intervals I_t , I_{box} and I_{ext} enter the computation as single balls whose endpoints are the exact rational interval endpoints; consequently every derived quantity is an enclosure valid simultaneously for all (t, y) in the corresponding rectangle, and its directed upper (resp. lower) endpoint bounds the supremum (resp. infimum) over that hull. More precisely, for each precision $p \in \{256, 512\}$ let $\text{Up}_p(E)$ denote the exact dyadic upper endpoint returned by Arb's directed upper-endpoint extraction for the p -bit ball evaluation of an expression E over I_t . That run uses

$$\widehat{\delta}_p = \text{Up}_p(\delta(N_*, I_t)), \quad \widehat{k}_p = \text{Up}_p(k(N_*, I_t)),$$

which satisfy $\widehat{\delta}_p \geq \widehat{\delta}$ and $\widehat{k}_p \geq \widehat{k}$. It substitutes these dyadic values for the exact maxima and extracts point lower bounds for σ_1, σ_2 over each rectangle. This is conservative throughout: increasing either frozen defect can only lower the exponents, and every subsequent occurrence of σ_i is through $u^{-\sigma_i}$ or $E_{t, \sigma_i}(u)$ with $u \geq 1$, both nonincreasing in σ_i .

The remaining inequalities. (1) The cap-validity conditions (28), in the equivalent form $\sigma_i > \frac{t}{2} \log N_*$, on both rectangles. (2) The cutoff conditions (31) for exactly the left endpoints listed there: $a = \lfloor M/d \rfloor$ for each $d \in \mathcal{S}$ at σ_1 , $a = M$ at σ_2 , and $a = M_{\text{err}}$ at both σ_1 and σ_2 , each with the t -factor evaluated at $\min I_t$ and the σ -factor at its lower bound over I_{ext} . (3) The A -side slope condition (32). (4) The hypotheses of Lemmas 5.5–5.9: $\sigma_1, \sigma_2 > 1$ on I_{ext} , $N_*(1 - \frac{t}{16N_*^2}) > 1$, $X_{N_*}(t) > 200$, $\log(N_*^2 - \frac{t}{16}) > 2$, and the sign conditions making $\Delta(N, t)$ and the $e_{C,0}$ quotient decreasing. (5) The exact rational identities: the mollifier normalization $\sum_d |\lambda_d| = 9.918746$, $\lambda_1 = 1$, $t_0 + y_0^2/2 = \frac{893927}{5 \cdot 10^6}$, and the interval inclusions $y_0 \in I_{\text{box}}$, $\max I_{\text{box}} \leq \sqrt{1-2t}$ for all $t \in I_t$, and $\max I_{\text{ext}} \geq \sqrt{1-2t_0}$, all checked in exact integer arithmetic. (6) The quantity D is assembled per (27) from an exact Dirichlet convolution $c_m(t)$ computed by direct double summation over (d, k) , and the four displayed constants are established as strict ball comparisons against exact rationals:

$$D < 0.999721, \quad M_{\max} < 1.608290, \quad E_{AB} + E_C < 1.1672 \times 10^{-8},$$

$$\frac{1 - D}{M_{\max}} - (E_{AB} + E_C) > 0.00017352.$$

The enclosures obtained at 512 bits include $D \leq 0.9997209094$, $M_{\max} \leq 1.6082891450$, $E_{AB} + E_C \leq 1.16716035 \times 10^{-8}$, and margin ≥ 0.0001735209 . Both precision runs prove all of the displayed inequalities. Since the intervals contain the entire stated domain, the positive lower bound proves the proposition. $\square$

Proposition 5.11 (Nonvanishing on the infinite tail). *For every $t \in I_t$, $y \in [y_0, \sqrt{1-2t}]$ and $x \geq X_{N_*}(t)$,*

$$|f_t(x + iy)| > e_A + e_B + e_{C,0},$$

and consequently $H_t(x + iy) \neq 0$ on this region.

Proof. Let N be the window index of x , so $N \geq N_*$. By Lemma 5.3, Lemma 5.8 and Proposition 5.10,

$$|f_t(x + iy)| \geq \frac{1 - D(N, t, y)}{M_{\max}(N, t, y)} > e_A + e_B + e_{C,0},$$

where the strict final comparison is precisely the direct interval inequality in Proposition 5.10, after applying both clauses of Lemma 5.8 and Lemma 5.9. Here $[y_0, \sqrt{1-2t}] \subseteq I_{\text{ext}}$. The point (x, y, t) lies in the region (10), so [8, Corollary 1.4] gives $H_t(x + iy) \neq 0$. $\square$

6 Hypothesis (iii): the barrier

6.1 Statement, and reduction to a closed rectangle

Throughout this section let

$$R = [X, X + 1] + i \left[\frac{1809}{10000}, 1 \right].$$

The goal of this section is the following proposition, proved after the boundary calculation in Section 6.5.

Proposition 6.1 (Nonvanishing on the barrier rectangle). *$H_t(z) \neq 0$ for all $z \in R$ and $0 \leq t \leq t_0$.*

Proposition 6.1 implies hypothesis (iii) of Theorem 2.1: the curved barrier region of that hypothesis satisfies, horizontally, $X \leq x \leq X + \sqrt{1 - y_0^2} \leq X + 1$; and vertically, $y \geq \sqrt{y_0^2 + 2(t_0 - t)} \geq y_0 > \frac{1809}{10000}$ (indeed $y_0^2 - (\frac{1809}{10000})^2 = \frac{234599}{10^8} > 0$) and $y \leq \sqrt{1 - 2t} \leq 1$; hence it is contained in $R \times [0, t_0]$.

6.2 Uniform effective error on the box, including $t = 0$

Proposition 6.2 (Uniform error bound on the barrier box). *For all $x \in [X, X + 1]$ and $0 \leq t \leq t_0$, the window index satisfies $N = \lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \rfloor = 690988$ and $N^2 < x/(4\pi)$. Moreover, for all $z \in R$ and $0 < t \leq t_0$, the error terms of Theorem 1.3, bounded with the constants of Section 2.3, satisfy*

$$e_A + e_B + e_{C,0} < 0.000356523011600040 < 0.00125. \quad (33)$$

Proof. The [barrier-error program](#) encloses every quantity below with outward-rounded FLINT/Arb balls at 256 bits, starting from the exact inputs X , $t_0 = \frac{129}{800}$, $y_{\min} = \frac{1809}{10000}$, and $N = 690988$.

Window-index constancy. The quantity $q(x, t) = \frac{x}{4\pi} + \frac{t}{16}$ is strictly increasing in each of x and t , and $N = \lfloor \sqrt{q} \rfloor$ is nondecreasing in q ; so it suffices to evaluate N at the two extreme corners $(X, 0)$ and $(X + 1, t_0)$ of the box. At both corners the outward enclosure of $\sqrt{q}$ lies in $[690988, 690989]$, so the floor is 690988 throughout the box. The same calculation proves

$N^2 < X/(4\pi) \leq x/(4\pi)$, and the equivalent inequality $2 \log N < L_{\min} := \log \frac{X}{4\pi}$; write also $L_{\max} := \log \frac{X+1}{4\pi}$.

Error bound. With N constant, for $0 < t \leq t_0$ the error terms are bounded by equation (23) of [8] and by (19); the program encloses the following majorants, each uniform on the box because every t - and y -dependent factor is evaluated at its worst endpoint: (i) by (17), using $(\cdot)_+ \leq 1 + 8/X^2$ and $y \geq y_{\min}$, $\operatorname{Re} s_* \geq a + \frac{t}{4}L_{\min} - \delta$ with $a = \frac{1+y_{\min}}{2}$ and $\delta = \frac{t_0(1+8/X^2)}{2X^2}$; since $\log n \leq \log N < \frac{1}{2}L_{\min}$, this gives $b_t(n) n^{-\operatorname{Re} s_*} \leq n^{-a_{\text{eff}}}$ with $a_{\text{eff}} = a - \delta$, uniformly in $t \in [0, t_0]$ (the t -dependent parts cancel against $\frac{t}{4} \log^2 n$, and $0 < a_{\text{eff}} < 1$); (ii) hence, by comparison with $\int_1^N u^{-a_{\text{eff}}} du$, $\sum_{n=1}^N b_t(n) n^{-\operatorname{Re} s_*} \leq 1 + \frac{N^{1-a_{\text{eff}}}-1}{1-a_{\text{eff}}}$; (iii) since $n^2 \leq N^2 < x/(4\pi)$, the bounds (16) and (18) give $|\gamma|n^y \leq e^{0.02y} \left(\frac{4\pi n^2}{x}\right)^{y/2} \leq e^{0.02}$ and $N^{|\kappa|} \leq \exp(\frac{t_0 \log N}{2(X-6)})$, so the prefactor of (23) is at most $1 + e^{0.02} \exp(\frac{t_0 \log N}{2(X-6)})$; (iv) since $0 < \log \frac{x}{4\pi n^2} \leq L_{\max}$, the last factor of (23) is at most $\exp(\frac{(t_0^2/16)L_{\max}^2 + 0.626}{X-6.66}) - 1$; (v) in (19), the prefactor is maximized at $(x, y) = (X, y_{\min})$, the factor $e^{-\frac{t}{16} \log^2 \frac{x}{4\pi}}$ is at most 1, $3^y + 3^{-y} \leq 3 + \frac{1}{3} = \frac{10}{3}$ (increasing in y , and $y \leq 1$), and the final quotient is bounded by maximizing its numerator ($L \leq L_{\max}$) and minimizing its denominator ($x \geq X$) separately. The program multiplies and sums these enclosures — yielding $e_A + e_B \leq 3.64 \times 10^{-10}$ and $e_{C,0} \leq 0.000356522648$ — and establishes the two strict inequalities of (33), the sharp constant being compared as the exact rational $356523011600040 \times 10^{-18}$. This proves the proposition uniformly on the box. $\square$

The constancy of N makes f_t a single analytic expression on R . The round number $0.00125 = \frac{1}{800}$ is the *approximation allowance* used by the prism criterion below; (33) says it is safely conservative.

Theorem 1.3 is stated in [8] for $t > 0$; the following lemma extends its conclusion to the closed endpoint $t = 0$ on the compact box.

Lemma 6.3 (Extension to $t = 0$). *For all $z \in R$ and all $0 \leq t \leq t_0$ (including $t = 0$), $B_t(z) \neq 0$ and*

$$|g_t(z) - f_t(z)| \leq 0.00125,$$

where $g_t = H_t/B_t$ and f_t is the sum (12) with the constant window index $N = 690988$.

Proof. For $0 < t \leq t_0$ this is Theorem 1.3 together with Proposition 6.2. For the endpoint, fix $z \in R$ and argue by a limit $t_j \downarrow 0$: (1) by dominated convergence in (1) (majorant $e^{t_0 u^2 + u} |\Phi(u)|$, integrable by the super-exponential decay of Φ), $H_t(z)$ is jointly continuous down to $t = 0$ on R ; (2) $B_t^{\pm 1}$ is continuous and $B_t \neq 0$ on $R \times [0, t_0]$, because the argument $\frac{1+y-ix}{2}$ of M_0 stays away from the branch cut (Section 2.2) and the deforming exponential never vanishes; (3) since the window index is constant, f_t is a fixed finite sum of functions continuous in t ; (4) the error majorant is continuous in t and bounded by 0.00125 on $(0, t_0]$ by (33). Passing to the limit in $|g_{t_j} - f_{t_j}| \leq 0.00125$ gives the claim at $t = 0$. $\square$

6.3 Uniform derivative bounds on the box

The moving-mesh argument below needs uniform bounds for $|\partial_z f_t|$ and $|\partial_t f_t|$ on R . The pointwise input is the following result; recall that by Proposition 6.2 the window index is constantly $N = 690988$ on the box, so that f_t is the fixed finite sum (12) with this N , jointly smooth in (z, t) , and the lemma applies to it verbatim.

Lemma 6.4 ([8, Lemma 8.4]). *In the region (10), and away from the jump discontinuities of N ,*

$$\begin{aligned} \left| \frac{\partial f_t}{\partial z} \right| &\leq \sum_{n=1}^N \frac{b_t(n)}{n^{\operatorname{Re} s_*}} \left(\frac{\log n}{2} + \frac{t \log n}{4(x-6)} \right) \\ &\quad + |\gamma| N^{|\kappa|} \sum_{n=1}^N \frac{b_t(n) n^y}{n^{\operatorname{Re} s_*}} \left(\frac{t \log n}{4(x-6)} + \left(\log \frac{|1+y+ix|}{4\pi} + \pi + \frac{3}{x} \right) \left(\frac{1}{2} + \frac{t}{4(x-6)} \right) \right), \end{aligned}$$

and

$$\begin{aligned} \left| \frac{\partial f_t}{\partial t} \right| &\leq \sum_{n=1}^N \frac{b_t(n)}{n^{\operatorname{Re} s_*}} \left(\frac{1}{4} \log n \log \frac{x}{4\pi n} + \frac{\pi}{8} \log n + \frac{2 \log n}{x-6} \right) \\ &\quad + |\gamma| N^{|\kappa|} \sum_{n=1}^N \frac{b_t(n) n^y}{n^{\operatorname{Re} s_*}} \left(\frac{1}{4} \log n \log \frac{x}{4\pi n} + \frac{\pi}{8} \log n + \frac{2 \log n}{x-6} \right. \\ &\quad \left. + \frac{1}{4} \left(\frac{\pi}{2} + \frac{8}{x-6} \right) \left(\log \frac{x}{4\pi} + \frac{8}{x-6} \right) \right). \end{aligned}$$

To convert these pointwise bounds into uniform ones we introduce the following frozen quantities. Write $y_{\min} = \frac{1809}{10000}$ (the bottom of R), $q(x) = \frac{x}{4\pi}$, and, as in (17),

$$h(x, y) = 1 - 3y + \frac{4y(1+y)}{x^2}, \quad c(x, y) = \frac{1}{4} \log q(x) - \frac{h(x, y)_+}{2x^2}, \quad c_0 = c(X, y_{\min}),$$

so that (17) reads $\operatorname{Re} s_* \geq \frac{1+y}{2} + t c(x, y)$. Define further

$$\begin{aligned} a(t) &= e^{0.02 y_{\min}} q(X)^{-y_{\min}/2} N^{t y_{\min}/(2(X-6))}, \\ P_t(u) &= u^{-\frac{1+y_{\min}}{2} + t(\frac{1}{4} \log u - c_0)}, \\ \theta(t) &= \frac{1}{2} + \frac{t}{4(X-6)}, \\ \Lambda_z(t) &= \left(\log \frac{|2+i(X+1)|}{4\pi} + \pi + \frac{3}{X} \right) \theta(t), \\ \Lambda_t &= \frac{1}{4} \left(\frac{\pi}{2} + \frac{8}{X-6} \right) \left(\log \frac{X+1}{4\pi} + \frac{8}{X-6} \right), \\ C_2 &= \frac{1}{4} \log \frac{X+1}{4\pi} + \frac{\pi}{8} + \frac{2}{X-6}, \end{aligned}$$

and the two summand majorants

$$\begin{aligned} F_z(u; t) &= P_t(u) \left[\theta(t) \log u + a(t) u^{y_{\min}} \left(\frac{t \log u}{4(X-6)} + \Lambda_z(t) \right) \right], \\ F_t(u; t) &= P_t(u) \left[(1 + a(t) u^{y_{\min}}) \log u \left(C_2 - \frac{\log u}{4} \right) + a(t) u^{y_{\min}} \Lambda_t \right], \end{aligned}$$

and finally

$$D_z(t) = \sum_{n=1}^{16} F_z(n; t) + \int_{16}^N F_z(u; t) du, \quad D_t(t) = \sum_{n=1}^{16} F_t(n; t) + \int_{16}^N F_t(u; t) du.$$

Two tight inequalities at the exact barrier parameters are needed below. The first, $N^2 < q(X)$, is part of Proposition 6.2. The second we record as a standalone statement.

Proposition 6.5 (Tail-exponent inequality). *At $X = 6000000185827$, $N = 690988$, $y_{\min} = \frac{1809}{10000}$,*

$$\frac{1}{4} \log \frac{q(X)}{N^2} > \frac{h(X, y_{\min})_+}{2X^2}.$$

Proof. The [tail-exponent program](#) encloses both sides with outward-rounded Arb balls at 256 bits, starting from the exact rational parameters. The computed enclosures are

$$\begin{aligned} \frac{1}{4} \log \frac{q(X)}{N^2} &= 2.2405813008 \dots \times 10^{-7} \pm 10^{-31}, \\ \frac{h(X, y_{\min})_+}{2X^2} &= 6.3513884954 \dots \times 10^{-27} \pm 10^{-51}. \end{aligned}$$

These intervals are disjoint, with the first lying above the second, and hence prove the strict inequality. $\square$

We shall use the following strict inequalities at the exact barrier parameters.

Proposition 6.6 (Derivative-bound conditions). *At $X = 6000000185827$, $N = 690988$, $y_{\min} = \frac{1809}{10000}$, $t_0 = \frac{129}{800}$:*

$$(C1) \quad X^2 > 8 \text{ and } \frac{1}{4(X+1)} - \frac{2}{X^3} > 0;$$

$$(C2) \quad 0.02 - \frac{1}{2} \log q(X) + \frac{t_0 \log N}{2(X-6)} + \frac{1}{2} \log N < 0;$$

$$(C3) \quad 2 \log N < \log q(X);$$

$$(C4) \quad \frac{12}{X^2} < 3;$$

$$(C5) \quad c_0 > \frac{1}{2} \log N;$$

$$(C6) \quad \frac{1-y_{\min}}{2} > \frac{1}{\log 16};$$

$$(C7) \quad C_2 > \frac{1}{4} \log N.$$

Proof. (C1), (C4) and (C6) are immediate by inspection ($X \approx 6 \times 10^{12}$; $\frac{1}{\log 16} = 0.3607 \dots < 0.40955 = \frac{1-y_{\min}}{2}$). For (C2), rewrite the left side as $0.02 + \frac{1}{2} \log \frac{4\pi N}{X} + \frac{t_0 \log N}{2(X-6)}$: here $\frac{4\pi N}{X} < \frac{8.8 \times 10^6}{6 \times 10^{12}} < 2 \times 10^{-6} < e^{-13}$, so the middle term is less than $-\frac{13}{2}$, while the last term is less than 10^{-12} ; hence the left side is less than $0.02 - 6.5 + 10^{-12} < 0$. (C7) follows from (C3): $C_2 - \frac{1}{4} \log N \geq \frac{1}{4} \log q(X) - \frac{1}{4} \log N > \frac{1}{4} \log N > 0$.

The remaining two conditions are exactly the statements established earlier: (C3) is the inequality $N^2 < q(X)$ of Proposition 6.2, and (C5) is Proposition 6.5, since $c_0 - \frac{1}{2} \log N = \frac{1}{4} \log \frac{q(X)}{N^2} - \frac{h(X, y_{\min})_+}{2X^2}$. $\square$

Lemma 6.7 (Pointwise domination on the box). *Assume (C1), (C2), (C4). Then for all $x \in [X, X+1]$, $y \in [y_{\min}, 1]$, $0 \leq t \leq t_0$ and $1 \leq n \leq N$, the n -th summand of the first (resp. second) bound of Lemma 6.4 is at most $F_z(n; t)$ (resp. $F_t(n; t)$). Consequently, for $0 \leq t \leq t_0$,*

$$\sup_{z \in R} \left| \frac{\partial f_t}{\partial z}(z) \right| \leq \sum_{n=1}^N F_z(n; t), \quad \sup_{z \in R} \left| \frac{\partial f_t}{\partial t}(z) \right| \leq \sum_{n=1}^N F_t(n; t).$$

Proof. Step 1: c is minimized at $(X, y_{\min})$. In y : $\partial_y h = -3 + \frac{4(1+2y)}{x^2} \leq -3 + \frac{12}{X^2} < 0$ by (C4), so h , hence h_+ , is nonincreasing in y and c is nondecreasing in y . In x : where $h > 0$,

$$\left| \partial_x \frac{h}{2x^2} \right| = \left| -\frac{4y(1+y)}{x^5} - \frac{h}{x^3} \right| \leq \frac{8}{x^5} + \frac{h_+}{x^3} \leq \frac{2}{x^3},$$

using $4y(1+y) \leq 8$, $8/x^2 \leq 1$ and $h_+ \leq 1$ (both from $X^2 > 8$; for $y > 0$, $h \leq 1$ reduces to $4(1+y) \leq 3x^2$, and at $y = 0$, $h = 1$). Where $h < 0$ the positive part is locally constant, and $h_+/(2x^2)$ is locally Lipschitz across the kink with the same bound. Hence for $x \in [X, X+1]$, $\partial_x c \geq \frac{1}{4x} - \frac{2}{X^3} \geq \frac{1}{4(X+1)} - \frac{2}{X^3} > 0$ almost everywhere by (C1), so c is increasing in x . Therefore $c(x, y) \geq c(X, y_{\min}) = c_0$ on the box.

Step 2: the B-side. By (17) and Step 1, $\operatorname{Re} s_* \geq \frac{1+y}{2} + t c(x, y) \geq \frac{1+y_{\min}}{2} + t c_0$, so

$$b_t(n) n^{-\operatorname{Re} s_*} \leq n^{\frac{t}{4} \log n - \frac{1+y_{\min}}{2} - t c_0} = P_t(n),$$

while the B-side brackets obey $\log n \left(\frac{1}{2} + \frac{t}{4(x-6)} \right) \leq \theta(t) \log n$ and

$$\log n \left(\frac{1}{4} \log \frac{x}{4\pi n} + \frac{\pi}{8} + \frac{2}{x-6} \right) = \log n \left(\frac{1}{4} \log q(x) - \frac{\log n}{4} + \frac{\pi}{8} + \frac{2}{x-6} \right) \leq \log n \left(C_2 - \frac{\log n}{4} \right),$$

since q is increasing and $\frac{1}{x-6}$ decreasing on $[X, X+1]$.

Step 3: the A-core. By (16)–(18), $|\gamma| N^{|\kappa|} n^y b_t(n) n^{-\operatorname{Re} s_*} \leq b_t(n) \mathcal{A}(x, y, t; n)$ with

$$\mathcal{A}(x, y, t; n) = e^{0.02y} q(x)^{-y/2} N^{\frac{ty}{2(x-6)}} n^{y - \frac{1+y}{2} - t c(x, y)}.$$

Using $\partial_y c \geq 0$ (Step 1) and $\log n \geq 0$,

$$\begin{aligned} \partial_y \log \mathcal{A} &= 0.02 - \frac{1}{2} \log q(x) + \frac{t \log N}{2(x-6)} + \frac{\log n}{2} - t (\partial_y c) \log n \\ &\leq 0.02 - \frac{1}{2} \log q(X) + \frac{t_0 \log N}{2(X-6)} + \frac{\log N}{2}, \end{aligned}$$

which is negative by (C2); so $\mathcal{A}$ is maximized at $y = y_{\min}$. At $y = y_{\min}$, freezing x conservatively ($q(x)^{-y_{\min}/2} \leq q(X)^{-y_{\min}/2}$, $N^{\frac{ty_{\min}}{2(x-6)}} \leq N^{\frac{ty_{\min}}{2(X-6)}}$, $c(x, y_{\min}) \geq c_0$) gives $b_t(n) \mathcal{A} \leq a(t) n^{y_{\min}} P_t(n)$.

Step 4: the A-side brackets. On $[X, X+1] \times [y_{\min}, 1]$, we have

$$\begin{aligned} \frac{t \log n}{4(x-6)} &\leq \frac{t \log n}{4(X-6)}, \\ \left(\log \frac{|1+y+ix|}{4\pi} + \pi + \frac{3}{x} \right) \left(\frac{1}{2} + \frac{t}{4(x-6)} \right) &\leq \left(\log \frac{|2+i(X+1)|}{4\pi} + \pi + \frac{3}{X} \right) \theta(t) = \Lambda_z(t), \end{aligned}$$

since $|1+y+ix| \leq |2+i(X+1)|$ (both coordinates increase) and $\frac{3}{x} \leq \frac{3}{X}$; and

$$\frac{1}{4} \left(\frac{\pi}{2} + \frac{8}{x-6} \right) \left(\log q(x) + \frac{8}{x-6} \right) \leq \Lambda_t,$$

each factor being bounded by its value at the appropriate endpoint.

Combining Steps 2–4 termwise bounds the two summands of Lemma 6.4 by $F_z(n; t)$ and $F_t(n; t)$. For $0 < t \leq t_0$ the displayed conclusions are then Lemma 6.4; at $t = 0$ they follow by continuity, as both sides of each inequality are continuous in t on the compact box (f_t being the fixed finite sum). $\square$

Lemma 6.8 (Head-and-tail evaluation). *Assume (C5), (C6), (C7). Then for each fixed $t \in [0, t_0]$ the functions $u \mapsto F_z(u; t)$ and $u \mapsto F_t(u; t)$ are positive and nonincreasing on $[16, N]$, and consequently*

$$\sum_{n=1}^N F_z(n; t) \leq D_z(t), \quad \sum_{n=1}^N F_t(n; t) \leq D_t(t).$$

Proof. Write $r = \log u \in [\log 16, \log N]$. The logarithmic derivative of P_t with respect to r is $p(r) = -\frac{1+y_{\min}}{2} + t(\frac{r}{2} - c_0) \leq -\frac{1+y_{\min}}{2}$, since $t \geq 0$ and $\frac{r}{2} \leq \frac{1}{2} \log N \leq c_0$ by (C5). Each of F_z, F_t is a nonnegative linear combination (with u -independent coefficients $\theta(t), \frac{t}{4(X-6)}, a(t), \Lambda_z(t), \Lambda_t$, all ≥ 0) of the five component functions

$$P_t(u)r, \quad P_t(u)u^{y_{\min}r}, \quad P_t(u)u^{y_{\min}}, \quad P_t(u)B(r), \quad P_t(u)u^{y_{\min}}B(r),$$

with $B(r) = r(C_2 - \frac{r}{4})$, which is positive on $[\log 16, \log N]$ by (C7). Their logarithmic r -derivatives are, respectively, at most

$$p + \frac{1}{r}, \quad p + y_{\min} + \frac{1}{r}, \quad p + y_{\min}, \quad p + \frac{B'}{B}, \quad p + y_{\min} + \frac{B'}{B},$$

and $\frac{B'}{B} = \frac{1}{r} - \frac{1}{4(C_2 - r/4)} \leq \frac{1}{r}$. Hence every component has logarithmic derivative at most

$$p + y_{\min} + \frac{1}{\log 16} \leq -\frac{1+y_{\min}}{2} + y_{\min} + \frac{1}{\log 16} = -\frac{1-y_{\min}}{2} + \frac{1}{\log 16} < 0$$

by (C6), so each component, and therefore each of F_z, F_t , is positive and decreasing on $[16, N]$. Finally, for decreasing positive F , $\sum_{n=17}^N F(n) \leq \int_{16}^N F(u) du$ (compare each term with the integral over $[n-1, n]$), which gives the displayed bounds. $\square$

Lemma 6.9 (Uniform derivative bounds). *Assume (C1)–(C7). Then for every $t \in [0, t_0]$,*

$$\sup_{z \in R} \left| \frac{\partial f_t}{\partial z}(z) \right| \leq D_z(t), \quad \sup_{z \in R} \left| \frac{\partial f_t}{\partial t}(z) \right| \leq D_t(t),$$

and hence, for every closed interval $[t_i, t_{i+1}] \subseteq [0, t_0]$, the relation

$$\sup_{z \in R, \tau \in [t_i, t_{i+1}]} |\partial_\tau f_\tau(z)| \leq \sup_{\tau \in [t_i, t_{i+1}]} D_t(\tau)$$

holds.

Proof. Combine Lemmas 6.7 and 6.8. $\square$

Remark 6.10. The program of Proposition 6.12 evaluates $D_z(t)$ and $D_t(t)$ with t an interval enclosure — for the prism bound, an enclosure of the whole interval $[t_i, t_{i+1}]$, which by inclusion isotonicity encloses $\sup_\tau D_t(\tau)$ — and computes the integrals $\int_{16}^N F(u) du$ by Arb's rigorous Gauss–Legendre integrator [2]. The integrands extend holomorphically in u away from the branch cut of the logarithm, and the calculation uses analytic logarithm and power operations on complex balls contained in this domain. The first sixteen summands are evaluated separately because Lemma 6.8 gives monotonicity only for $n \geq 16$; the remaining terms are bounded by the integral exactly as in that lemma.

6.4 The prism criterion

The time interval $[0, t_0]$ will be covered by finitely many closed “prisms” $R \times [t_i, t_{i+1}]$. The following lemma reduces nonvanishing of H on a prism to finitely many verifiable conditions at its initial time t_i .

Lemma 6.11 (Prism criterion). *Let $0 \leq t_i < t_{i+1} \leq t_0$, and let $z_1, \dots, z_m$ (cyclically ordered, $z_{m+1} = z_1$) be a mesh on ∂R with consecutive spacing at most h . Let $D_z \geq \sup_{z \in R} |\partial_z f_{t_i}(z)|$ and $D_t \geq \sup_{z \in R, \tau \in [t_i, t_{i+1}]} |\partial_\tau f_\tau(z)|$ (as supplied by Lemma 6.9), and suppose:*

(a) $|f_{t_i}(z_j)| \geq M_i$ for all j , where

$$M_i > \frac{D_z h}{2} + D_t (t_{i+1} - t_i) + 0.00125; \quad (34)$$

(b) using the clockwise convention in the computation, every argument increment satisfies

$$\arg(f_{t_i}(z_j)/f_{t_i}(z_{j+1})) \in (-\pi, \pi),$$

and their sum gives winding number 0 for the polygon $(f_{t_i}(z_j))_j$ about 0.

Then $H_\tau(z) \neq 0$ for all $z \in R$ and $\tau \in [t_i, t_{i+1}]$.

Proof. On each half of a mesh subedge, both the true image curve $F(s) = f_{t_i}(\text{subedge}(s))$ and its endpoint chord lie in the closed disk of radius $D_z h/2$ about the nearer endpoint value (this is the factor $\frac{1}{2}$ in (34)); each disk is convex, so the straight-line homotopy from the true boundary curve to the mesh polygon stays inside these disks. Moving to time $\tau \in [t_i, t_{i+1}]$ displaces values by at most $D_t(\tau - t_i)$, and passing from f_τ to $g_\tau = H_\tau/B_\tau$ by at most 0.00125 (Lemma 6.3). Condition (34) says every resulting disk, centered at a mesh value of modulus $\geq M_i$, excludes the origin; hence the entire three-stage homotopy (polygon $\rightarrow$ true f_{t_i} -boundary image $\rightarrow f_\tau(\partial R) \rightarrow g_\tau(\partial R)$) avoids zero. In particular each chord avoids the origin, so its argument variation in the implemented clockwise orientation is the principal argument of the stated endpoint ratio. The winding number of the polygon about the origin is a well-defined integer; by (b) it is 0, and the zero-avoiding homotopy gives winding number 0 for $g_\tau(\partial R)$ as well, for every $\tau \in [t_i, t_{i+1}]$. Since $B_\tau \neq 0$ and B_τ is holomorphic on a neighborhood of R (Lemma 6.3 and Section 2.2), g_τ is holomorphic there and has the same zeros as H_τ , and g_τ is zero-free on ∂R ; the argument principle then gives $H_\tau \neq 0$ on R . $\square$

6.5 The prism decomposition

The computational input of the barrier argument is the following.

Proposition 6.12 (Prism decomposition). *There exist times $0 = \tau_0 < \tau_1 < \dots < \tau_{883}$ with $\tau_{882} < t_0 \leq \tau_{883}$ such that for each $0 \leq i < 883$, the hypotheses (a)–(b) of Lemma 6.11 hold for the prism $R \times [\tau_i, \min(\tau_{i+1}, t_0)]$, with margin in (34) at least 0.5198498946 in every prism.*

Proof. The **barrier program** carries out the following three phases. It uses outward-rounded Arb balls, and every decisive inequality is strict.

(1) *Domain hypotheses and truncation.* At 192-bit precision the calculation proves the uniform expansion-parameter bound $|w| \leq 0.6480 < 0.66$ on the whole slab (via the shift

bounds $|\operatorname{Re}(\log n_0 - \alpha)| < 0.694$ and $|\operatorname{Im}(\log n_0 - \alpha)| < 0.7855$ for both α -arguments used below, where $n_0 = N/2$; the bound $0.0882 < 0.089$ for the modulus of the γ -type factor, both for its exact expression (on a 256×128 subdivision of the box) and for the implemented majorant; and, by a rigorous direct sum over $n \leq N$ with geometric tail factors, the complete two-variable 62-term Taylor truncation bound $1.9543 \times 10^{-22} < 10^{-20}$ — the allowance that is attached as an error ball to every boundary mesh value in phase (3).

(2) *Coefficient calculation.* Define the 62×62 complex matrix

$$a_{e,k} = \sum_{n=1}^N n^{\frac{i(X+\frac{1}{2})-1}{2}} \frac{(\log \frac{n}{n_0})^e}{e!} \frac{(\frac{1}{4} \log^2 \frac{n}{n_0})^k}{k!}$$

and enclose each entry by direct summation. A calculation with the two Taylor directions shows that the polynomial evaluation in phase (3) reproduces, exactly up to the truncation bound of phase (1), the two sums of (12) at the evaluation point, with the heat factor $b_t(n)$ emerging from the algebra of the two Taylor directions.

(3) *Covering the time interval.* Starting at $\tau_0 = 0$, at each prism start τ_i the calculation evaluates the derivative bounds $D_z(\tau_i)$ and (for the proposed next prism, with the time parameter interval-valued over the entire closed prism) D_t , by the head-plus-integral quadratures of Section 6.3, rounding both up to exact integers. It then meshes ∂R with $4\lceil D_z \rceil - 4$ points (spacing at most $h = 1/(\lceil D_z \rceil - 1)$), proves every mesh enclosure of f_{τ_i} to exclude zero, every argument increment to lie strictly in $(-\pi, \pi)$, and the per-rectangle winding enclosure to lie strictly in $(-\frac{1}{4}, \frac{1}{4})$, establishing winding number 0. It chooses τ_{i+1} adaptively at half the established slack of (34), as an exact dyadic number, and proves the inequality $\min_j |f_{\tau_i}(z_j)| > \frac{D_z h}{2} + D_t(\tau_{i+1} - \tau_i) + 0.00125$. Adjacent intervals have identical exact dyadic endpoints, the first time is exactly 0, the final endpoint satisfies $\tau_{883} \geq \frac{129}{800} = t_0$, and the aggregate winding enclosure lies strictly in $(-\frac{1}{4}, \frac{1}{4})$. (The final prism may extend slightly beyond t_0 ; restricting it to its intersection with $[0, t_0]$ only shrinks the displacement terms in (34), so the conclusion for the full final prism covers the truncated one, as required by the statement.)

The calculation produces 883 prisms, with aggregate winding enclosure $[\pm 1.18 \times 10^{-14}]$ and worst margin $0.51984989461387254365 \pm 2 \times 10^{-20}$, which lies above the exact rational threshold $5198498946 \times 10^{-10}$. The first time is exactly 0, consecutive intervals share exact dyadic endpoints, and the last endpoint is at least t_0 ; hence the 883 closed prisms cover the whole time interval. This proves the proposition. $\square$

Proof of Proposition 6.1. By Proposition 6.12 and Lemma 6.11, $H_\tau(z) \neq 0$ for $z \in R$ and $\tau \in [\tau_i, \min(\tau_{i+1}, t_0)]$, for each i . The union of these intervals is $[0, t_0]$. $\square$

7 Completion of the proof

Proof of Theorem 1.1. The parameters (4) satisfy $0 < t_0 < \frac{1}{2}$ and $0 < y_0^2 < 1 - 2t_0$. Hypothesis (i) of Theorem 2.1 holds by Section 3, since $X/2 < T_{PT}$. Hypothesis (ii) holds by combining Proposition 4.11 (nonvanishing of H_{t_0} for $x_* \leq x < x_{3840001}$, $y_0 \leq y \leq \sqrt{1 - 2t_0}$) with Proposition 5.11 applied at $t = t_0 \in I_t$ (nonvanishing for $x \geq x_{3840000} = X_{N_*}(t_0)$): the two ranges overlap on the full window $W_{3840000}$, so together they cover all $x \geq x_*$. Hypothesis (iii) holds by Proposition 6.1, since the curved barrier region is contained in $R \times [0, t_0]$ (Section 6).

Theorem 2.1 then gives

$$\Lambda \leq t_0 + \frac{y_0^2}{2} = \frac{893927}{5000000} = 0.1787854. \quad \square$$

A Programs used in the proof

The calculations in Propositions 4.3, 4.9, 4.10, 5.10, 6.2, 6.5, and 6.12 use exact integer or rational inputs and outward-rounded FLINT/Arb arithmetic. The source links in Table 2 point to Git commit 6222740efae58d773a70ff929cf12a34932d2af0. Each program returns a mathematical inequality only when the relevant intervals are disjoint; an overlapping interval is reported as unresolved.

Result	Source	Calculation
Prop. 4.3	Program P4.3	Evaluates L_N for every $690988 \leq N \leq 3840000$ and proves $L_N \geq 791366 \times 10^{-12}$.
Prop. 4.9	Program P4.9	Covers the four (L, y) domains by closed rectangles and proves $Gh_+ < \frac{L}{2} C_0 $ on every rectangle.
Prop. 4.10	Program P4.10	Computes the five finite-region error majorants at 256 and 512 bits and proves $e_A + e_B + e_{C,0} < E_{\max} < 234 \times 10^{-9}$.
Prop. 5.10	Program P5.10	Evaluates the cutoff contraction and error terms on $I_t \times I_{\text{box}}$ and proves a post-error margin greater than 0.00017352.
Prop. 6.2	Program P6.2	Proves that the window index is constant on the barrier box and that the effective-approximation error is less than 0.00125.
Prop. 6.5	Program P6.5	Encloses both sides of the tail-exponent inequality and proves that their difference is positive.
Prop. 6.12	Program P6.12	Builds the Taylor coefficients, covers $[0, t_0]$ by 883 closed prisms, and proves zero winding with a minimum margin greater than 0.5198498946.

Table 2: Programs used for the finite calculations. The displayed labels are direct links to the corresponding source files in the [project repository](#).

The 256- and 512-bit outputs for Program P4.10 are available as [the 256-bit output](#) and [the 512-bit output](#).

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